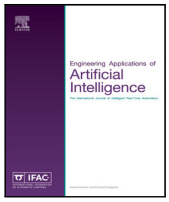
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Survey paper



A scientometric analysis of quantum driven innovations in intelligent transportation systems

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ABSTRACT

Intelligent transportation systems have witnessed a transformative evolution through the integration of quantum technology. The current paper delves into the contemporary landscape of research progress and challenges within the domain of Quantum Enhanced Intelligent Transportation Systems (QEITS). The study employs scientometric methods to comprehensively analyze the developmental status, research distribution, and future trends of QEITS. The research utilized data from the Web of Science and Scopus databases. The study explores publication patterns, keyword co-occurrence networks, document co-citation network analyses, and country trends in various domains of QEITS such as "dynamic vehicle routing", "advanced vehicle connectivity and data security", "smart transportation management and facilities", and "electric vehicle technology and sustainability". These analyses unveil dominant research areas such as quantum evolutionary algorithms, quantum cryptography, smart parking, and electric vehicle energy management. The findings provide insights into the intellectual landscape and current research trends, nurturing innovation, and facilitating well-informed decision-making within QEITS to the researchers and academic community.

1. Introduction

Intelligent Transportation Systems (ITS) have been under development since the early 1970s, representing the forthcoming trajectory of transportation systems. ITS integrates cutting-edge technologies such as electronic sensors, data transmission, and intelligent control mechanisms into transportation systems (Zhang et al., 2023; Tchappi et al., 2020). Technologies such as Information and Communications Technology (ICT), Artificial Intelligence (AI), Big Data Analytics, Communication Technologies, and Cloud Computing are playing a pivotal role in the ITS. ICT refers to technologies used for gathering, storing, retrieving, processing, analyzing, and transmitting information. It includes both hardware and software systems, as well as networks and services, that enable these functions. The primary objective of ITS revolves around enhancing services for both drivers and passengers within transportation systems (Zhu et al., 2018), leveraging advancements in AI to achieve this goal. AI serves as a cornerstone in ITS, enabling sophisticated functions such as traffic management, autonomous vehicle navigation, predictive maintenance, and real-time decision-making. By harnessing AI capabilities, ITS functions as a cohesive framework that deploys a diverse array of communication, control, vehicle sensing, and electronics technologies (Rani and Sharma, 2023) to efficiently oversee and regulate traffic dynamics, alleviate congestion, offer optimal travel routes and boost overall system efficiency. Moreover, AI-driven ITS endeavors to ensure road safety, enhance driving comfort, and disseminate up-to-date road information to users (Singh and Gupta, 2015; Azad et al., 2022). Over the recent decades, ITS has emerged as a potent approach for enhancing vehicle performance on roadways. Within the realm of ITS, data can be sourced from different channels, including smart cards, GPS (Global Positioning System), sensors, video detectors, and even social media platforms. Employing precise and efficient data analytics techniques on this disorganized data can yield enhanced services for ITS. As ITS advances, the volume of generated data has evolved from the Terabyte scale to the Petabyte scale (Zhang et al., 2011). Big Data analytics introduces a novel technical approach to ITS. However, it continues to possess the subsequent constraints:

(i) Achieving real-time navigation for expansive road networks poses a challenge due to the sluggish pace of big data transmission and processing.
(ii) Security concerns arise from vulnerabilities in the software code and parameters of road systems.
(iii) The accurate identification of exceptional and specialized events remains an issue, leading to delays in traffic management and a deficiency in reliability.

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Considering that the coherent superposition in quantum mechanics enables parallel computation and exponential growth, quantum entanglement and the no-cloning theorem provide an assurance of complete data security (Cheung et al., 2019). Furthermore, the quantum tunneling effect enhances the likelihood of escaping local extremes to attain the globally optimal solution, along with other advantageous features (Finnila et al., 1994; Banerjee and Majhi, 2008). Consequently, the integration of Quantum Computing (QC) into ITS aims to accelerate data processing, ensure data security and reinforce overall system dependability (Tengilimoglu et al., 2023). Therefore, the focus is directed towards unveiling the potential of the Quantum-Enhanced Intelligent Transportation System (QEITS). The paper analyzed the QEITS by categorizing it into four distinct domains based on the literature (Harwood et al., 2021; Sood et al., 2021; Yi et al., 2022; Cheong et al., 2023; Elvas and Ferreira, 2021):

(i) Dynamic Vehicle Routing and Navigation (DVRN): It involves the optimization of real-time vehicle routes, considering factors such as traffic congestion and demand fluctuations. QC implementations in DVRN promise rapid calculation of optimal routes, leveraging parallel processing to adjust routes dynamically, thereby enhancing fleet efficiency and reducing travel time.
(ii) Advanced Vehicle Connectivity and Data Security (AVCDS): This category ensures seamless vehicle communication while prioritizing data security. Quantum advancements, including post-quantum cryptography, fortify data protection and connectivity reliability, ensuring robust encryption and enhanced communication protocols for vehicle networks.
(iii) Smart Transportation Management and Facilities (STMF): This category encompasses exploring methods to improve the management and infrastructure of transportation systems through the incorporation of quantum technologies. After implementing QC, communication, and other advancements, it aims to enhance the efficiency, reliability, and security of transportation networks.
(iv) Electric Vehicle Technology and Sustainability (EVTS): It focuses to advance electric vehicle technology and sustainability by integrating quantum-inspired innovations to optimize battery performance, charging infrastructure, and energy management. Through these solutions, QEITS enhances the efficiency and sustainability of Electric Vehicles (EVs), fostering a cleaner transportation ecosystem.

To access a visual representation of the QEITS research field, Fig. 1 provides the taxonomy, which offers a comprehensive depiction of its intricate components (Wang et al., 2021; Zhu et al., 2019; Osaba et al., 2022; Shim, 2021; Dey et al., 2014; Vianna et al., 2004; Elvas and Ferreira, 2021). The proposed work delves into the four aforementioned domains through the execution of analyses encompassing publication trends, co-citation patterns, and co-occurrence networks.

1.1. Background

ITS represent a transformative approach to modernizing transportation infrastructure by leveraging advanced technologies to improve traffic management, safety, and efficiency. Unlike typical IoT networks, ITS is characterized by its stringent real-time requirements, high mobility, large data volumes, and critical safety needs (Wang et al., 2021). These unique features necessitate specialized network designs that standard IoT platforms cannot adequately support. ITS applications, such as autonomous driving, traffic signal control, vehicle-to-everything (V2X) communication, and emergency response systems, demand extremely low latency and real-time data processing to ensure timely decision-making, prevent accidents, and maintain traffic flow and safety (Alam et al., 2016). The high mobility of vehicles within ITS results in dynamic network topologies, requiring frequent handovers and robust communication protocols that can maintain seamless connectivity and reliable data exchange across rapidly changing network environments. Additionally, ITS generates vast amounts of heterogeneous data, including traffic conditions, environmental data, vehicle diagnostics, and pedestrian movements, which require scalable and efficient data management solutions that can process and analyze data in real-time to enable adaptive and predictive responses (Kaffash et al., 2021).

Security is another paramount concern in ITS, given its direct impact on human safety and critical infrastructure. Traditional IoT security mechanisms are often insufficient to address the sophisticated threats facing ITS, such as cyber-attacks on autonomous vehicles, traffic control systems, and V2X communications. Ensuring the integrity, confidentiality, and availability of data in ITS is crucial as any breach can lead to severe consequences, including accidents and traffic disruptions. The unique characteristics of ITS significantly impact the design and implementation of quantum networks (Jabbar et al., 2022). Standard quantum communication protocols and platforms cannot be directly applied to ITS due to several critical factors, including latency sensitivity, dynamic network management needs, and heightened security concerns. Quantum networks must be designed to optimize protocols for real-time performance to meet the stringent latency requirements of ITS applications. Traditional IoT solutions are not equipped to handle the high mobility and frequent topology changes inherent in ITS, necessitating the development of adaptive quantum communication protocols that can dynamically adjust to changing network conditions and maintain continuous, reliable communication links (Mohammed, 2023).

Enhanced Quantum Key Distribution (QKD) techniques are essential to ensure secure communications in the highly mobile and dynamic ITS environment. These techniques must be tailored to handle the fast-moving and frequently changing nodes typical of vehicular networks, ensuring robust security even during handovers and rapid topology shifts. The integration of QKD with classical cryptographic methods can provide layered security, enhancing the overall robustness of ITS against diverse cyber threats. Moreover, achieving seamless integration between quantum and classical communication networks is crucial to leveraging existing ITS infrastructure effectively while introducing the benefits of quantum communication, such as unparalleled security and efficient data transfer rates. This integration involves developing hybrid architectures that can seamlessly switch between quantum and classical communication channels as needed, ensuring continuity and reliability in all scenarios, including urban, suburban, and rural environments (Mohammed, 2023; Stavdas et al., 2024).

Furthermore, the design of quantum networks for ITS must consider the scalability of communication protocols to accommodate the increasing number of connected vehicles and infrastructure components. This involves developing quantum repeaters and advanced error correction methods to maintain signal integrity over long distances and through complex urban landscapes. Additionally, the deployment of quantum networks in ITS must be cost-effective and energy-efficient to ensure practical implementation and sustainability (Lv and Shang, 2023). Quantum networks also need to support high data throughput to manage the large volume of data generated by ITS applications without compromising performance. In addressing why standard platforms are insufficient for ITS, it is crucial to consider the particular modifications and enhancements required for quantum networks to effectively support ITS environments (Granelli et al., 2022). Adaptive quantum communication protocols must be developed to handle the dynamic and high-speed nature of vehicular networks. These protocols should be capable of providing real-time responses and maintaining high levels of reliability and security despite frequent changes in network topology. Enhanced QKD techniques are necessary to secure data transmission against potential eavesdropping and cyber threats, ensuring the privacy and integrity of sensitive information such as vehicle location and traffic data (Hasan et al., 2023).

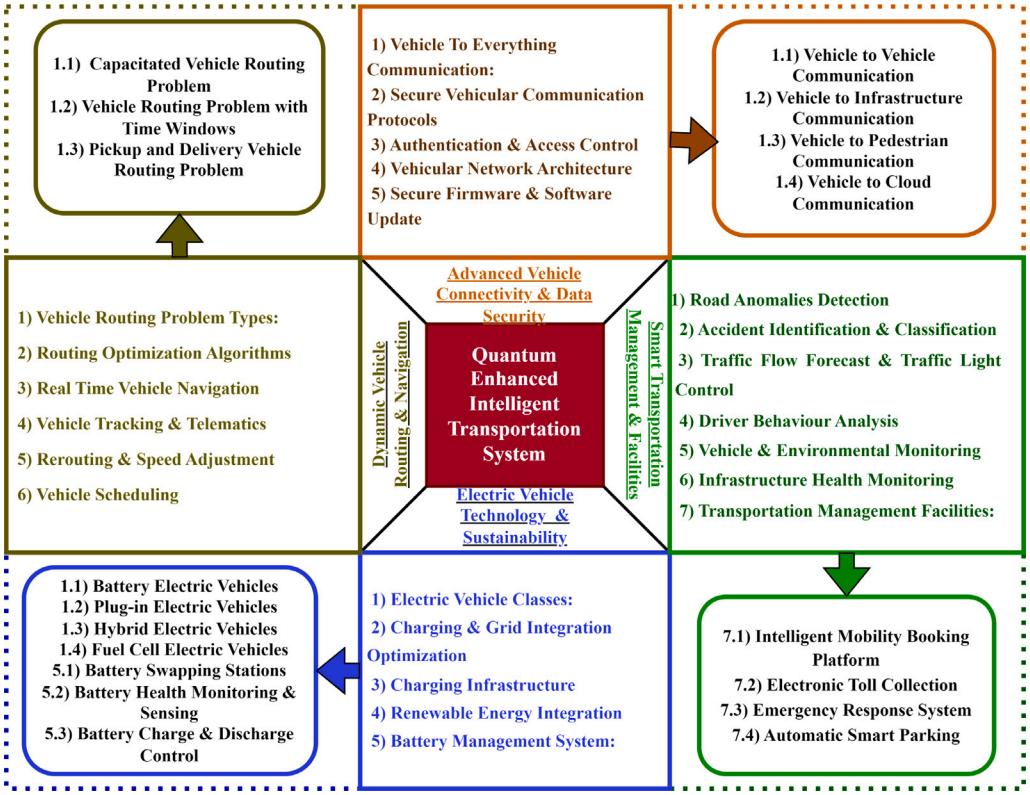


Fig. 1. QEITS taxonomy.

Moreover, the seamless integration of quantum and classical networks involves creating hybrid systems that can leverage the strengths of both technologies. For instance, quantum networks can be used for securing critical communications, while classical networks can handle less sensitive data and provide redundancy. This hybrid approach ensures that the overall system is robust, flexible, and capable of meeting the diverse needs of ITS applications. The implementation of such integrated systems requires careful planning and the establishment of new standards and protocols to ensure interoperability and efficient resource management (Majumder et al., 2021; Borah and Bitragunta, 2021).

1.2. Motivation

In recent years, the intersection of QC and ITS has garnered increasing attention from both researchers and practitioners. QC with its potential to solve complex optimization and simulation problems exponentially faster than classical computers, presents a transformative opportunity for enhancing the efficiency, safety, and sustainability of transportation networks (Wang et al., 2021). Challenges in ITS include data integration complexities, cybersecurity, scalability, regulatory hurdles, user acceptance, environmental impact, technological integration, and innovation demands (Chattopadhyay et al., 2021). Addressing these requires interdisciplinary efforts and the utilization of advanced Quantum Algorithms (QAs) for robust, secure, and sustainable transportation solutions. The integration of QAs and technologies in ITS holds promise for improving traffic flow management, optimizing route planning, enhancing vehicle-to-infrastructure communication, and advancing autonomous vehicle capabilities. QAs such as Quantum Annealing, Quantum Machine Learning, Quantum Optimization like "Quantum Approximate Optimization Algorithm" (QAOA), and Quantum Cryptography can be utilized to overcome challenges in ITS (Cooper, 2021). However, despite the growing interest in this interdisciplinary field, there remains a need for a comprehensive scientometric analysis to systematically map out and evaluate the evolution, trends, and impacts of quantum-driven innovations within ITS. This scientometric review aims to explore the current landscape of quantum-driven innovations in ITS, examining methodologies, applications, and emerging trends. By synthesizing existing literature and identifying open challenges, this article aims to pave the way for future research directions that fully leverage QC's potential to transform transportation systems worldwide.

1.3. Paper organization

The current study is organized into ten sections to present the scientometric analysis of QEITS. Section 2 provides the discussion on applications of QC technology in ITS. Section 3 discusses the related work. Section 4 provides the methodology used for data collection and the selection of tools for data processing. Section 5 presents the quantitative and qualitative analyses in terms of publications temporal analysis, geographical distribution, DCoN and KCoN analysis. The overall discussion is provided in Section 6, challenges and future research directions are discussed in Section 7. Furthermore, implications of the current study are described in Section 8 and the last Section 9 outlines the conclusions of the study.

2. Implications of QC in ITS

QC stands as a groundbreaking computational paradigm, harnessing the power of quantum superposition, entanglement, and coherence to enable quantum parallel processing (Bennett and Shor, 1998; Castelvecchi, 2017; Bhatia and Sood, 2020). Furthermore, a detailed discussion on the fundamentals or phenomena of QC through which ITS can be enhanced is provided below.

Table 1
Popular qubits and involved companies.

Qubit type	Definition	Involved companies
"Superconducting Qubits"	"A superconducting qubit, constructed from superconducting materials, represents the quantum states through variations in its electrical properties, such as current or voltage."	Rigetti, Google, IBM, OQC, QuTech
"Trapped Ion Qubits"	"Trapped ion qubits are quantum units formed by confining ions using electromagnetic fields, encoding quantum states via specific ion energy levels."	IONQ, Universal Quantum, QUANTINUUM, AQT
"Photonic Qubits"	"Photonic qubits are quantum units encoded in photons, exploiting properties like polarization or phase for quantum information processing."	XANADU, ORCA Computing, PsiQuantum
"Neutral Atoms"	"Neutral atoms, with balanced protons and electrons, are used as qubits in QC, manipulated via electromagnetic fields and lasers."	QuEra, atom computing, PASQAL, ColdQuanta
"Quantum Dots"	"Quantum dots are nanoscale semiconductor particles with confined electrons, serve as potential qubits for QC."	Intel, QUANTUM MOTION, Silon Quantum Computing, diraq

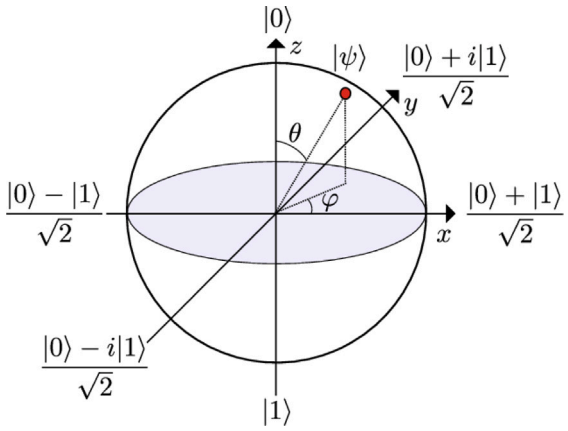


Fig. 2. Representation of Bloch Sphere (Kockum and Nori, 2019).

Fundamentals of QC

(i) Q-Bit: A qubit, the basic building block of quantum computing, is implemented within a quantum computer (Datta et al., 2008) being mathematically depicted by a unit vector within a two-dimensional Hilbert space, which is a complex vector space where quantum states reside (Brunner et al., 2008). In this context, a qubit's state is symbolized using Dirac notation, expressed in Eq. (1).

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

where α and β are the probability amplitudes of the states defined using complex numbers, satisfy the subsequent condition: $|\alpha|^2 + |\beta|^2 = 1$
Further, the geometrical representation of Bloch Sphere is shown in Fig. 2 and the type of qubits and involved companies are presented in Table 1 for a comprehensive understanding of qubit.

(ii) Superposition: Superposition enables a system to exist in numerous states concurrently. In a 2-qubit system, this yields four states: 00, 01, 10, and 11. Quantum computers use this property to compute across all states concurrently, achieving exponential computational capacity growth through parallelism (N qubits represent 2^N classical bits) (Friedman et al., 2000). The visual view of superposition is shown in Fig. 3 for better understanding.

(iii) Entanglement: Quantum entanglement is a phenomenon where the states of particles become correlated, regardless of distance, such that measuring one instantaneously affects the other (Horodecki et al., 2009). Fig. 3 provides a visual representation of entanglement between two qubits, aiding comprehension.

(iv) Q-Bit Register: A qubit register, also referred to as a qubit individual, is a collection of qubits into a cohesive quantum entity (Bradley et al., 2019). For example, an "n-qubit" register is defined as: $|\xi\rangle = \Sigma i \in (0,1)^n \xi|i\rangle = \Sigma i_1 i_2 \ldots . i_n \in (0,1)\xi i_1 i_2 \ldots . i_n |i_1 i_2 \ldots . i_n\rangle$, here the complex number $\xi i_1 i_2 \ldots . i_n$ satisfy the normalization requirement of $\Sigma i \in (0,1)^n |\xi|^2 = 1$ (Bhatia et al., 2019).

(v) Q-Gates: Quantum gates, denoted by the reversible unitary operator X, are employed to alter the state of qubits. They adhere to the requirement $X^T X = X X^T$, where X^T signifies the Hermitian Adjoint of X. The details of quantum gates are shown in Fig. 4.

QC holds significant potential to enhance various applications in the context of ITS, including infrastructure management, data network optimization, encryption security, evolutionary computation, etc. Furthermore, Fig. 5 provides the pictorial representation of QC and ITS integration clusters and a detailed discussion is provided below:

2.1. Infrastructure management

QC offers transformative potential in enhancing the infrastructure of ITS. Leveraging QAs such as quantum annealing (Yarkoni et al., 2022), QAOA (Azad et al., 2022), and "Quantum Machine Learning" techniques like "Quantum Support Vector Machines" (Ding et al., 2021), quantum technology can address various challenges and optimize operations across multiple facets of transportation infrastructure. By applying these algorithms, QC can rapidly detect accidents, road anomalies, and traffic patterns, enabling quicker response and improved safety measures. Additionally, quantum neural networks (Abbas et al., 2021) and quantum convolutional networks can analyze traffic camera footage and sensor data to extract meaningful features and patterns, facilitating more accurate and efficient analysis of traffic flow and congestion. Quantum clustering algorithms, such as quantum k-means (Poggiali et al., 2024) or quantum spectral clustering, can group similar traffic patterns or road conditions, providing insights into areas requiring targeted infrastructure improvements or maintenance. Moreover, utilizing quantum reinforcement learning (Wauters et al., 2020) algorithms can optimize traffic signal control strategies, while quantum optimization algorithms can optimize resource allocation in transportation infrastructure projects, ensuring optimal utilization of budget and resources. By harnessing the computational power and advanced algorithms of QC, ITS infrastructure can be revolutionized, ushering in a new era of optimized, efficient, and safe transportation systems.

2.2. Data network

QC offers promising avenues for enhancing data processing and analysis within vehicular networks, leading to improved communication and decision-making capabilities. By harnessing its capability to manage extensive datasets and execute intricate calculations at unprecedented speeds, quantum computers enable real-time data analytics for vehicle-to-vehicle and vehicle-to-infrastructure communication. Incorporating advanced QAs and techniques, such as quantum machine

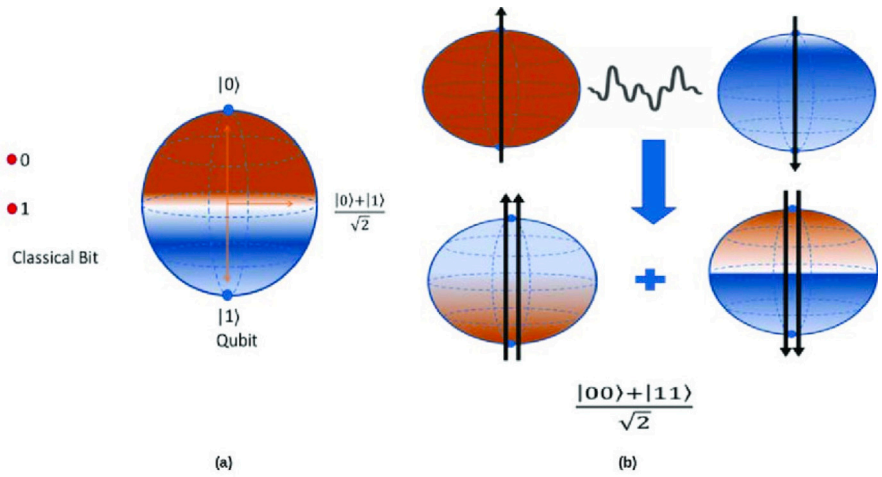


Fig. 3. Representation of superposition and entanglement phenomenons (Maheshwari et al., 2022).

Gate Name	Notation	Matrix Representation
Hadamard	H	$\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$
Pauli-X	X	$\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$
Pauli-Y	Y	$\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}$
Pauli-Z	Z	$\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$
Phase	S	$\begin{bmatrix}1 & 0\\ 0 & i\end{bmatrix}$
Controlled-not		$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{bmatrix}$
Swap		$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$
Controlled-Z	= Z	$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -1\end{bmatrix}$
Controlled-phase	S	$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & i\end{bmatrix}$

Fig. 4. Matrix representations of quantum gates.

learning algorithms and quantum data compression, can significantly enhance the data network's performance (Xu et al., 2023). Data compression techniques such as quantum singular value decomposition or quantum principal component analysis. "Quantum Machine Learning Algorithms", such as "Quantum Support Vector Machines" and "Quantum Neural Networks", can efficiently extract insights from the extensive data generated within vehicular networks (Schuld et al., 2014). These algorithms enable faster decision-making and facilitate predictive maintenance strategies, ultimately enhancing the overall efficiency of transportation systems. Additionally, quantum data compression techniques can minimize bandwidth requirements and latency, ensuring efficient data exchange in real-time (Yu et al., 2019). By compressing data using QAs, vehicular networks can reduce the transmission overhead and optimize resource utilization. Furthermore, QC can improve the analysis of traffic patterns, road conditions, and vehicle performance data, leading to better traffic management and optimization. "Quantum optimization algorithms", such as "quantum annealing" and "QAOA", can optimize the routing and allocation of data packets within vehicular networks (Ilarri et al., 2015). By efficiently solving complex optimization problems, these algorithms reduce congestion, minimize latency, and enhance overall network performance.

2.3. Encryption algorithms and communication security

QC presents a dual landscape of challenges and opportunities for encryption in vehicular networks. Traditional encryption algorithms like "Rivest–Shamir–Adleman" (RSA) and "Elliptic Curve Cryptography" (ECC) face potential vulnerabilities from quantum computers by utilizing the Shor algorithm, which excels at efficiently factoring large numbers and solving discrete logarithm problems (Ugwuishiwu et al., 2020). However, the emergence of quantum-resistant encryption techniques, such as "Lattice-based Cryptography", "Code-based Cryptography", and "Hash-based Cryptography", offers promising avenues for securing communication channels (Nejatollahi et al., 2019). Moreover, QC can expedite the advancement of quantum key distribution protocols like "Bennett-Brassard" 1984 (BB84) and Ekert 1991 (E91), bolstering the security of vehicle-to-vehicle and vehicle-to-infrastructure communications against eavesdropping and tampering. Utilizing post-quantum cryptography involves the development of encryption techniques that withstand potential threats posed by quantum computers, further strengthening the security of vehicular networks (Sutradhar, 2023). In this way, while QC presents challenges to traditional encryption, it also catalyzes the development of robust encryption solutions, ensuring the integrity and confidentiality of data exchanged within vehicular networks.

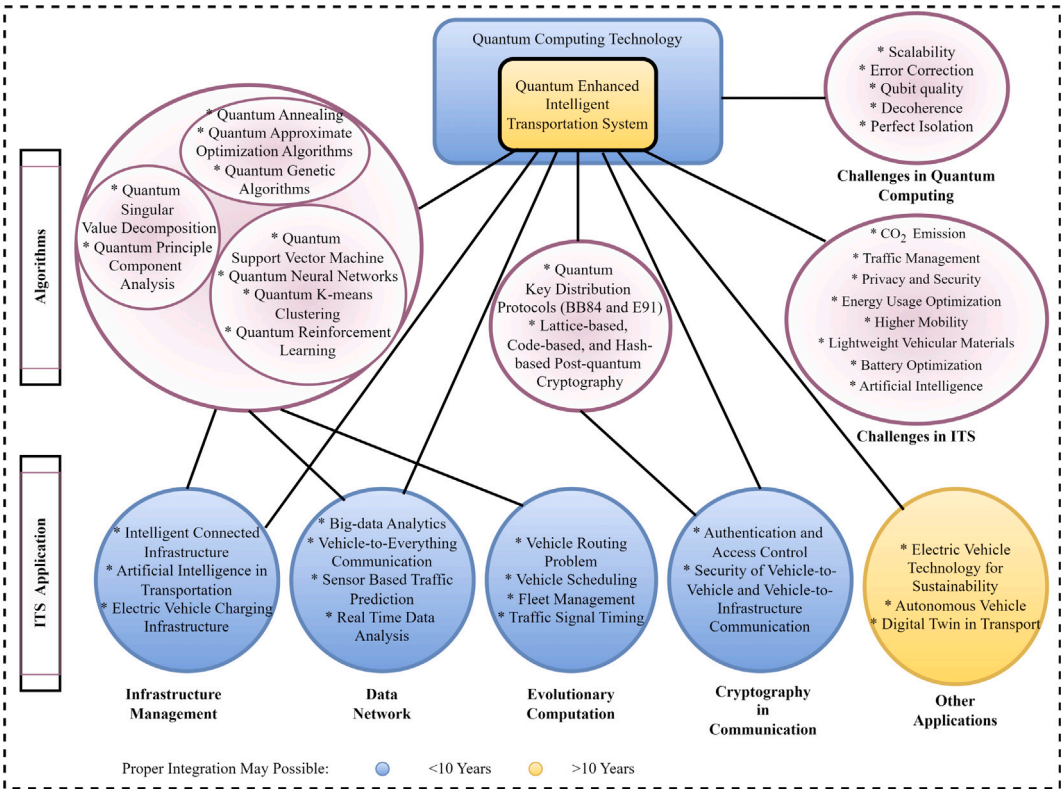


Fig. 5. Pictorial representation of QC and ITS integration clusters.

2.4. Evolutionary computation

Evolutionary computation, deeply rooted in principles of natural selection and genetics, offers a diverse suite of optimization techniques for solving complex problems within ITS. These methodologies, such as genetic algorithms, genetic programming, evolutionary strategies, and particle swarm optimization, have proven invaluable for enhancing various transportation operations specifically vehicle route optimization problems (Prins, 2004). They also play a crucial role in traffic signal timing, vehicle scheduling, fleet management, infrastructure design, traffic prediction, and forecasting. However, the advent of QC introduces new possibilities for enhancing evolutionary computation and other metaheuristic algorithms in ITS. Quantum-inspired evolutionary algorithms, such as "Quantum Genetic Algorithms" (QGA) (Lin and Tang, 2021), leverage principles of quantum mechanics to explore solution spaces more efficiently and overcome computational bottlenecks. By harnessing quantum principles like superposition and entanglement, these algorithms can potentially provide superior optimization performance and scalability, particularly for large-scale transportation system complexities (Wang et al., 2012). Additionally, parallelized evolutionary computing paradigms empowered by QC hardware offer the potential for accelerated optimization processes, further advancing the efficiency and resilience of ITS operations. Integrating QC principles into evolutionary computation methodologies represents a promising frontier for achieving heightened optimization, efficiency, and adaptability in managing transportation operations within ITS.

2.5. Advancement in EVs

QC presents a transformative opportunity for advancing EVs by addressing critical challenges in energy storage, battery optimization, and overall vehicle efficiency (Sanguesa et al., 2021). Its unparalleled processing power and capacity to handle complex optimization problems offer unprecedented avenues for innovation in the EV industry (Dalyac et al., 2021). Leveraging QAs such as quantum annealing, variational quantum eigensolvers, and quantum machine learning, researchers can optimize battery materials, electrode designs, and charging algorithms to enhance energy density, charging speed, and battery lifespan in EVs (Luckow et al., 2021). Moreover, QC facilitates precise simulations of molecular structures and chemical reactions, leading to the discovery of novel materials for next-generation batteries and energy storage systems. By adopting a path that integrates quantum-inspired optimization algorithms, quantum simulation techniques, and machine learning models, research efforts in the EV field can be accelerated significantly. Additionally, applying quantum-enhanced optimization techniques for supply chain management, vehicle routing, and energy distribution can optimize the entire lifecycle of EVs. Embracing QC methodologies and fostering interdisciplinary collaborations will unlock new horizons for sustainable transportation, contributing to a cleaner, and greener future.

2.6. Time estimations for QC integration

Regarding time estimations about QC integration in these applications, it is challenging to provide precise predictions. QC is in its nascent phase, encountering numerous technical hurdles including scalability, error correction, qubit quality, decoherence, achieving perfect isolation, and designing chip architecture (Sood and Monika, 2023) needs to overcome before it becomes widely applicable. Progress in hardware development, error correction, and algorithm design will influence the timeline for integration into various applications. Expectations of gradual advancements in these areas over the next decade are reasonable. According to the Gartner Hype Cycle, 2018 (Hype Cycle for Emerging Technologies, 2018) and Gill et al. (2022), QC technology will take 5–10 years to reach the plateau. However, quantum machine learning and post-quantum cryptography will take the next 4–5 years. Therefore, early implementations may emerge in specialized domains where quantum advantages are most pronounced, such as optimization problems in vehicle routing optimization, communication, and traffic management. However, widespread adoption across ITS, EVs, energy

management, and related domains may take longer, depending on factors such as technological development, research breakthroughs, and practical scalability challenges. Although, in this NISQ era, Hybrid Algorithms are used to enhance this knowledge domain. Overall, QC is poised to revolutionize ITS and associated applications, offering unprecedented opportunities for optimization, efficiency, and innovation in urban mobility and transportation infrastructure.

3. Related work

QC has sparked breakthroughs in various domains, including transportation systems. QEITS has gained prominence for its impressive algorithmic performance, leading to a surge in scientific literature. This section reviews relevant QEITS literature, offering a concise summary of findings and identifying the research gap. QC technologies are increasingly enhancing the field of ITS. Some studies that showcase quantum solutions are described ahead. As traffic flow management is a critical issue in ITS, Neukart et al. (2017) explored how quantum annealing algorithms leveraging D-Wave Systems Quantum Processing Units (QPUs) to optimize traffic flow in dense road networks. It discusses using a hybrid quantum–classical approach due to current QPU limitations, focusing on real-time car position data redistribution. Future QPU advancements are anticipated to handle larger and more complex problems, potentially revolutionizing real-time transportation optimization. On the other hand, Zhang et al. (2020) utilized the QGA and Learning Vector Quantization (LVQ) Neural Network to enhance prediction traffic network prediction accuracy and convergence speed. Experimental results across various datasets show that QGA-LVQ outperforms traditional methods like genetic algorithms, and wavelet neural networks, achieving significantly lower Mean Absolute Percentage Error and demonstrating stability for real-time traffic prediction and management in smart city environments.

Moreover, Hu et al. (2018) proposed an urban traffic light scheduling model based on "Quantum Particle Swarm Optimization Algorithm" (QPSOA) and performed better than other standard algorithms like Dynamic Scheduling Algorithm (DSA), Genetic Algorithm (GA), and Comprehensive Learning Particle Swarm Optimization (CLPSO). QAs are also proving advantageous in various domains such as vehicle routing, communication, and EV energy management, where they demonstrate superior performance and results. In this context, Quantum Support Vector Machine (Mohanty et al., 2024), QAOA (Azad et al., 2022), and Quantum Annealing Methods (Feld et al., 2019) are providing better results in solving vehicle routing problems. For vehicular communication and security, quantum and post-quantum protocols are being utilized to enhance encryption, ensure data integrity, and improve communication reliability. Sutradhar and Venkatesh (2024) introduced a quantum protocol for secure vehicular communication aimed at preventing collisions on foggy highways. Leveraging Nash equilibrium, the protocol addresses vulnerabilities to quantum computer attacks and minimizes computation costs with an O(1) complexity for secret signal sharing by the Traffic Control System (TCS). Regular updates of secret signals enable vehicles to compute them collaboratively, ensuring robust communication and enhancing safety measures on foggy highways. In contrast, Yadav et al. (2024) proposed a quantum-safe blockchain framework for V2V communication in autonomous vehicles. It ensures authenticity using lattice-based techniques like Module Learning With Errors and Module Short Integer Solution, resistant to quantum attacks with high throughput and low latency. The framework integrates quantum-safe signature aggregation and a blockchain to enhance safety, traffic flow efficiency, and fuel conservation in vehicular networks. However, both studies are providing the quantum solutions for the communication perspective. Furthermore, it has been observed that Quantum Particle Swarm Optimization (QPSO) (Lihao et al., 2016) and Quantum Artificial Bee Colony (Cai et al., 2019) algorithms are beneficial for managing energy in EVs.

Furthermore, several review articles have been found, Wang et al. (2021) explored QC's potential in enhancing ITS for better path planning, management, and facility layout. While universal quantum computers are in the early stages, algorithm improvements are pursued. Special-purpose quantum computers like D-Wave show promise in practical ITS applications. Osaba et al. (2022) provided an 18-year literature review of QC integration with routing problems by analyzing 53 papers. The study found a trend towards practical articles in recent years. Challenges include tackling real-world problems, understanding quantum hardware limitations, and proper problem parameterization, suggesting a promising future for QC in solving routing challenges. Cooper (2021) investigated applying QC to transportation planning, discussing potential benefits and challenges. QAs are explored for network analysis, shortest path computation, etc. However, challenges arise in implementing reversible computation due to space constraints, requiring further research to realize the potential of quantum transportation algorithms. Shim (2021) addressed the threat of quantum computers and explored post-quantum alternatives, emphasizing vehicular communications as a priority application. The authors surveyed post-quantum signature schemes, delving into theoretical foundations, implementation details, and tradeoffs for practical use. The study discusses challenges such as larger key sizes and performance tradeoffs, urging further research to ensure secure communication in the post-quantum era.

Additionally, in the context of the scientometric review, there are many comprehensive studies are conducted on ITS. Scientometric review on ITS for EVs (Elvas and Ferreira, 2021), this study highlighted the need for significant charging infrastructure expansion in the European Union to accommodate 30 million EVs by 2030. Key challenges include preventing grid strain, ensuring comprehensive coverage, and developing efficient energy management. Recent literature emphasizes optimization, standardization, and regulatory improvements, underscoring the importance of ongoing research and infrastructure development for sustainable EV growth. A study by Sood et al. (2021) analyzed the evolution of vehicular communication, from basic signaling to advanced Vehicle-to-Everything (V2X) technologies, using a scientometric approach on publications from 2009 to 2020. Data were processed using Comma-separated values files and VOSViewer for keyword analysis and clustering. Findings indicate significant research areas and evolving technologies, routing protocols, data communication standards, traffic control, security, and privacy. The study's comprehensive analysis highlights future research directions in blockchain, cloud computing, and challenges with EVs. Another scientometric study by Liu et al. (2021) analyzed 1536 bibliographic records from the Web of Science, focusing on traffic forecasting research. It identified emerging trends such as neural networks, deep learning, and spatio-temporal correlations are noted. The study offered insights into influential institutions, scholars, and future research directions in traffic forecasting. Singh and Gupta (2015) explored, how ITS addresses global traffic congestion using advanced technologies. While well-established in developed nations, ITS adoption in India, Brazil, China, and South Africa is recent. The study focuses on four components: Advanced Traveler Information System, Advanced Traffic Management System, Advanced Public Transportation System, and Emergency Management System, reviewing architectures, identifying knowledge gaps, and proposing future research directions. Further, Seth et al. (2022) identified that the Internet of Vehicles (IoV) enhances vehicular communication for improved safety and efficiency. It reviews IoV architectures, focusing on seven-layer models, and evaluates advanced Medium Access Control and routing protocols suitable for various environments. Key challenges like security, scalability, and ensuring reliable service are highlighted. The study emphasizes the need for ongoing advancements in IoV to enhance automotive safety, efficiency, and connectivity. A study by Emadi et al. (2008) explored electric, hybrid electric, and plug-in hybrid vehicles to meet emission reduction and fuel efficiency goals. It emphasizes the role of power electronics in supporting higher electrical loads in vehicle

Table 2
State-of-the-art comparison analysis in ITS.

Reference	Approach	Research theme	Quantum computing integration	AT	DWC	PG	PC	IS	ERA	RP
Wang et al. (2021)	Systematic	ITS	✓	X	X	X	X	X	X	✓
Osaba et al. (2022)	Bibliometric	Routing Problems	✓	X	X	✓	✓	✓	X	X
Cooper (2021)	Systematic	Transportation Modeling	✓	X	✓	X	X	✓	X	X
Shim (2021)	Systematic	Secure Vehicular Communication	✓	X	✓	X	X	✓	X	X
Elvas and Ferreira (2021)	Scientometric	EVs	X	✓	✓	X	X	X	✓	X
Sood et al. (2021)	Scientometric	Distributed Vehicular Networks	X	✓	X	✓	✓	✓	✓	X
Liu et al. (2021)	Scientometric	Traffic Forecasting	X	✓	X	✓	✓	✓	✓	X
Najmi et al. (2017)	Scientometric	Transport Domain	X	✓	X	X	X	✓	✓	X
Singh and Gupta (2015)	Systematic	ITS	X	X	✓	X	X	X	✓	X
Seth et al. (2022)	Systematic	Internet of Vehicles	X	X	✓	X	X	✓	X	✓
Emadi et al. (2008)	Systematic	Power Electronics and Motor Drivers in EVs	X	X	✓	X	X	✓	X	X
Deng et al. (2023)	Systematic	6G Autonomous ITS	X	X	✓	X	X	✓	X	✓
Jabbar et al. (2022)	Systematic	Blockchain Technology for ITS	X	X	✓	X	X	✓	X	✓
Shaaban et al. (2021)	Systematic	Visible Light Communication for ITS	X	X	X	X	X	✓	X	✓
Proposed work	Scientometric	QEITS	✓	✓	✓	✓	✓	✓	✓	✓

designs. Key points include current trends, future strategies, challenges like weight, cost, and customer value, and ongoing efforts to enhance power electronics and motor drive technologies across EV platforms.

Furthermore, Deng et al. (2023) explored the evolution of ITS towards 6G Autonomous ITS (6G-AITS). They emphasize enhancing transportation efficiency through automation and advanced connectivity. Key goals include defining the requirements for 6G-AITS and highlighting features like high intelligence, security, and sustainability. Proposed applications include integrated space–air-ground networks, vehicle infrastructure cooperation, and urban air mobility. A study by Jabbar et al. (2022) showed Blockchain technology's role in ITS and IoV. It traces Blockchain's evolution from pre-Bitcoin to Blockchain 2.0, highlighting innovations such as Ethereum and smart contracts. It classified research according to IoV layers (processing, communication, security) and outlines current trends, challenges, and future research directions. The survey concludes with a comparison to previous literature and identifies remaining open issues in Blockchain for IoV. A review on Visible light Communication (VLC) for ITS by Shaaban et al. (2021) examined the dual function of data transmission and illumination. It explores VLC's potential in ITS, identifies implementation barriers, and suggests future research directions. Further, Najmi et al. (2017) provided bibliometric analysis on 35,712 journal articles from top transport journals to track trends in transport planning and management. It identifies major themes like transportation theory, economics, network analysis, and safety through co-keyword and journal co-citation networks. Key findings highlight a surge in research output since 2010. Future research could expand qualitative analysis, study scholar networks, explore interdisciplinary links, and validate findings using broader databases.

For comparative analysis, Table 2 shows the comparison of studies existing in the ITS based on ten parameters namely Approach, QC Integration, Research Topic, Analytical Tool (AT), Domain-wise Categorization (DWC), Publication Growth (PG), Prolific Countries (PC), Intellectual Structure (IS), Emerging Research Areas (ERA), Research Prospects (RP). The extant scholarly literature within the QEITS domain demonstrates that there are a limited number of studies that present the knowledge structure in QEITS using the scientometric technique. Moreover, prominent articles across diverse sub-domains employing scientometric, bibliometric, and systematic methodologies frequently exhibit incomplete utilization of the parameters delineated in the table. Therefore, the current work focuses on addressing all constraints outlined in the table from a quantum perspective. The scientometric approach in this study illustrates the growth of publications, prominent countries, Document Co-citation Network (DCoN), and Keyword Co-occurrence Network (KCoN) to identify current status, emerging topics, and future research directions.

3.1. Research questions

The objective of the investigation is to offer a comprehensive understanding of the QEITS scientific literature. This is accomplished by addressing the following key research questions:

RQ1: How do publication patterns evolve over time in QEITS research?

RQ2: Which nations actively engage in research for specific QEITS domains?

RQ3: What is the intellectual structure of distinct QEITS domains, and what are the emerging topics and trends that define research trajectories?

RQ4: What challenges, opportunities for future research, and critical insights exist within this realm of knowledge?

3.2. Contributions

The study offers a scientometric analysis of QEITS literature, spanning from its inception to 2023, to construct a roadmap for forthcoming research endeavors. It makes significant contributions to the QEITS knowledge domain presented in Fig. 6 and written as follows:

(i) The current research work provides a scientometric review of the QEITS research field, aiding new researchers in building their knowledge base.
(ii) It offers a taxonomy of concepts and subdivisions of the QEITS knowledge domain to enhance breadth and depth.
(iii) The current study categorized the QEITS research field into four significant domains (DVRN, AVCDS, STMF, and EVTS) and analyzed each domain comprehensively.
(iv) The proposed study provides a way to identify the intellectual landscape, influential topics, and most active research topics by employing DCoN and KCoN analysis on each identified domain.
(v) It presents the implications of QC in ITS, current research challenges, outlines future research directions, and contributes significantly to informed decision-making within academia.

4. Research methodology

The present research employs a scientometric methodology to investigate evolving paradigms and prospective trajectories in QEITS. Fig. 7 illustrates the methodological framework adopted to visualize and scrutinize the scholarly literature surrounding this field. QEITS research is categorized into four distinct categories labeled D1, D2, D3, and D4 in Fig. 7. The study utilized the Scopus and Web of Science (WoS) bibliometric databases to explore these categories. Initially, 2263 records were identified in the WoS database, distributed as follows: D1



Fig. 6. Key contributions of the proposed work.

= 395, D2 = 543, D3 = 487, and D4 = 838. In contrast, the Scopus database revealed 20,797 records, with distributions of D1 = 1907, D2 = 6025, D3 = 5529, and D4 = 7333. After applying the filtration criteria outlined in Fig. 7, a total of 369 records were retrieved from the WoS database, categorized as D1 = 84, D2 = 110, D3 = 101, and D4 = 82. The Scopus database yielded 651 records, distributed as D1 = 108, D2 = 207, D3 = 171, and D4 = 172. The limited number of records indicates the nascent stage of QC technology. Following the merging of datasets and the removal of duplicates using the CiteSpace visualization tool, a total of 1016 records were included for scientometric analysis, with category distributions of D1 = 185, D2 = 313, D3 = 267, and D4 = 251. Various analyses, including publication patterns, prominent country analysis, DCoN analysis, and KCoN analysis, have been conducted on this dataset to extract key insights. Finally, the research findings are synthesized, highlighting challenges and future research directions, with detailed descriptions provided in the sections below.

4.1. Database and analytical tool selection

The article centers on the examination of QEITS using the scientometric technique. Various databases, such as Scopus, WoS, IEEE Explore, Google Scholar, and Science Direct, offer access to scientific literature (Kaur and Bhatia, 2021; Chen and Song, 2019). It is noteworthy that Scopus and WoS databases are often employed interchangeably by researchers to conduct bibliometric studies (Sood et al., 2022b; Bakkalbasi et al., 2006). These databases, encompass a wide range of content and encompassing high-impact scientific journals. Therefore, Scopus and WoS databases are utilized in this study. Regarding the compilation of documents from both databases, the process of merging and subsequent elimination of duplicate entries was executed using the CiteSpace tool. Both databases encompass diverse attributes, including publication year, document type, research area, country, and language selections, thereby offering comprehensive insights into the dataset (Kaur et al., 2023; Sood and Agrewal, 2024). Moreover, their integrated analytical tools facilitate graphical and tabular visualization of analyses (Sood et al., 2022c; Sood and Monika, 2023).

Conducting scientometric analysis necessitates the use of data visualization tools. A spectrum of such tools is accessible for scholars, including CiteSpace (Chen, 2006), Gephi (Jacomy et al., 2014), VOSviewer (Van Eck and Waltman, 2010), SciMAT (Cobo et al., 2012), and UCInet (Johnson, 1987). However, in this study CiteSpace visualization tool was utilized which was created by Dr. C. Chen, affiliated with Drexel University, USA. CiteSpace serves as a professional scientometric and visualization tool, enabling structured and temporal analyses of scholarly contributions based on factors such as authors, institutions, references, keywords, and country categories (Sood et al., 2022a; Synnestvedt et al., 2005).

4.2. Data search strategy

The "Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA)" approach, proposed by Liberati et al. (2009) and Zaidan et al. (2022), is widely recognized and deemed suitable for evaluating scientific data. The current study utilized this approach to extract the dataset. The extraction of relevant records within each domain is achieved by utilizing the search queries through both WoS and Scopus databases. In the search field of each database, the respective significant keywords pertaining to each domain were entered.

(i) DVRN: "Quantum" AND ("Vehicle Routing OR Vehicle Navigation") NOT ("Unmanned OR Drones OR AVCDS category keywords OR STMF category keywords OR EVTS category keywords").
(ii) AVCDS: "Quantum" AND ("Vehicle Connectivity OR Vehicle Data Security OR VANET OR Vehicle Networking OR Internet of Vehicles OR vehicle Blockchain OR Vehicle Authentication") NOT ("Unmanned OR Drones OR DVRN category keywords OR STMF category keywords OR EVTS category keywords").

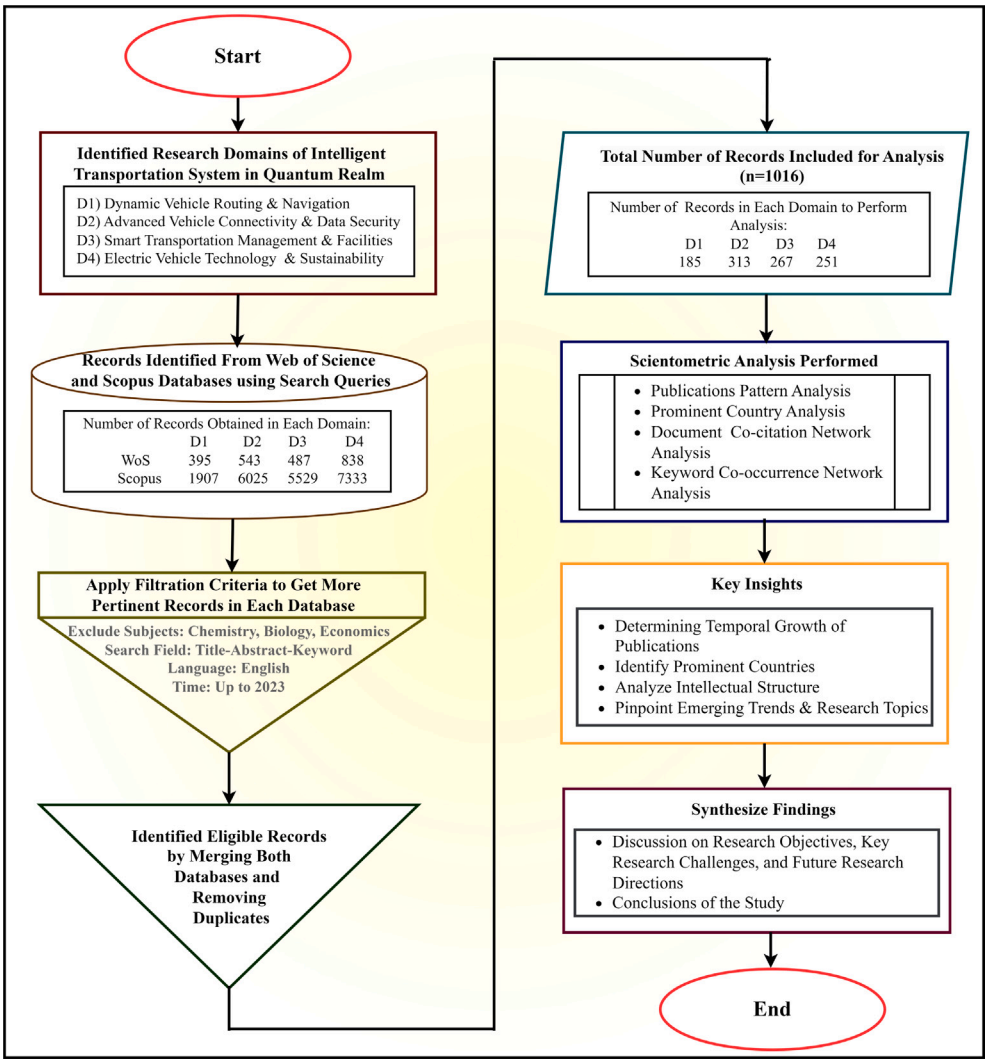


Fig. 7. Flow diagram of research methodology.

(iii) STMF: "Quantum" AND ("Vehicle Transportation Management OR Road Anomalies Detection OR Vehicle Degradation System OR Accident Identification and Classification OR Traffic Flow OR Traffic Light Control OR Driver Behavior Analysis OR Electronic Toll Collection OR Smart Parking OR Vehicle Monitoring") NOT ("Unmanned OR Drones OR Routing OR DVRN category keywords OR AVCDS category keywords OR EVTS category keywords").

(iv) EVTS: "Quantum" AND "Electric Vehicle" NOT ("Unmanned OR Drones OR DVRN category keywords OR AVCDS category keywords OR STMF category keywords").

5. Quantitative and qualitative analyses

5.1. Publications temporal analysis

The current section provides a comprehensive analysis of the publication trajectory in the QEITS research field, spanning from its inception to 2023. Fig. 8 visually represents the overall and category-wise publication patterns of QEITS literature extracted from the WoS and Scopus databases. By examining these patterns across four categories of quantum-enhanced intelligent systems, the analysis uncovers emerging trends, key developments, and areas of increased research activity. This insight helps researchers focus on cutting-edge advancements, prioritize their studies, and identify potential collaboration opportunities in these rapidly evolving fields. Additionally, the graphical visualization aids in swiftly understanding the domain's research dynamics and progression. In Fig. 8, the temporal dimension spans from 2000 to 2023 along the horizontal axis, while the vertical axis portrays the number of publications received within this temporal span. Within the graph, the prominent blue trend line with red fill illustrates the annual progression of Total Publications (TP) in the QEITS literature. The pattern depicts a gradual research progression, despite this decline occurring in 2007, 2010, and 2020. The major decline that has been done in 2020 is likely due to researchers shifting their focus to areas such as COVID-19. Subsequently, there is a pronounced resurgence after 2020. However, it is important to note that there is a decline in 2023, likely attributed to the data only being inclusive of up to August 2023. Furthermore, the other four lines with yellow, brown, blue, and green colors in the graph depict the publications counts in each domain of QEITS literature. The analysis illustrates a consistent increase in publications across all domains until 2002, followed by a substantial surge in publication count. Notably, the period from 2003 to 2019 demonstrates a high degree of unpredictability. Within the scientific literature pertaining to each category, the years 2019 to 2020 exhibit a decrease in publications for DVRN (13), AVCDS (31), STMF (17), and EVTS (11). The year 2022 emerges as noteworthy for registering the highest frequency of published papers. However, it is important to note that the domain of (DVRN) experienced a decrement in publication count during the years 2011 (1), 2013 (8), and 2016 (5). Furthermore, a discernible trend

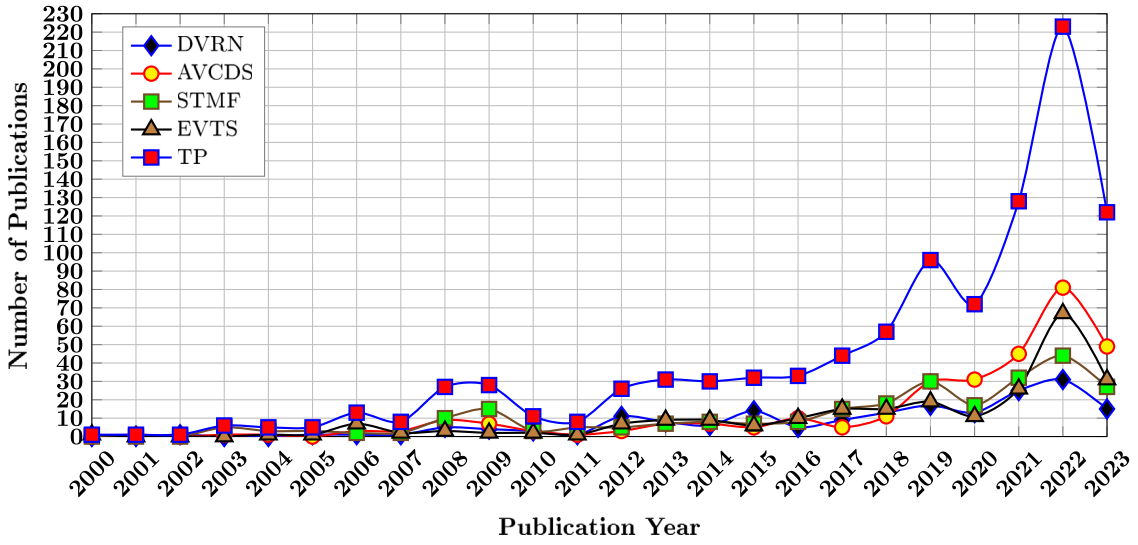


Fig. 8. Temporal distribution of publications in QEITS knowledge domain.

Table 3
Top productive nations in each QEITS domain.

QEITS domain	Country 1	Country 2	Country 3	Country 4	Country 5
DVRN	China(75, 21.37%)	USA(26, 16.05%)	India(16, 12.05%)	Japan(10, 22.73%)	UK(9, 20.45%)
AVCDS	China(97, 27.64%)	USA(56, 34.57%)	India(36, 0.31%)	Canada(10, 32.26%)	Italy(6, 19.35%)
STMF	China(83, 23.65%)	USA(29, 17.09%)	India(45, 35.16%)	Canada(10, 32.26%)	South Korea(8, 80.00%)
EVTS	China(96, 27.35%)	USA(35, 21.06%)	India(29, 22.66%)	Japan(15, 34.09%)	Germany(9, 45.00%)

has been observed in recent years, reflecting a substantial expansion in the publication counts across all categories within the realm of QEITS literature.

5.2. Geographical distribution

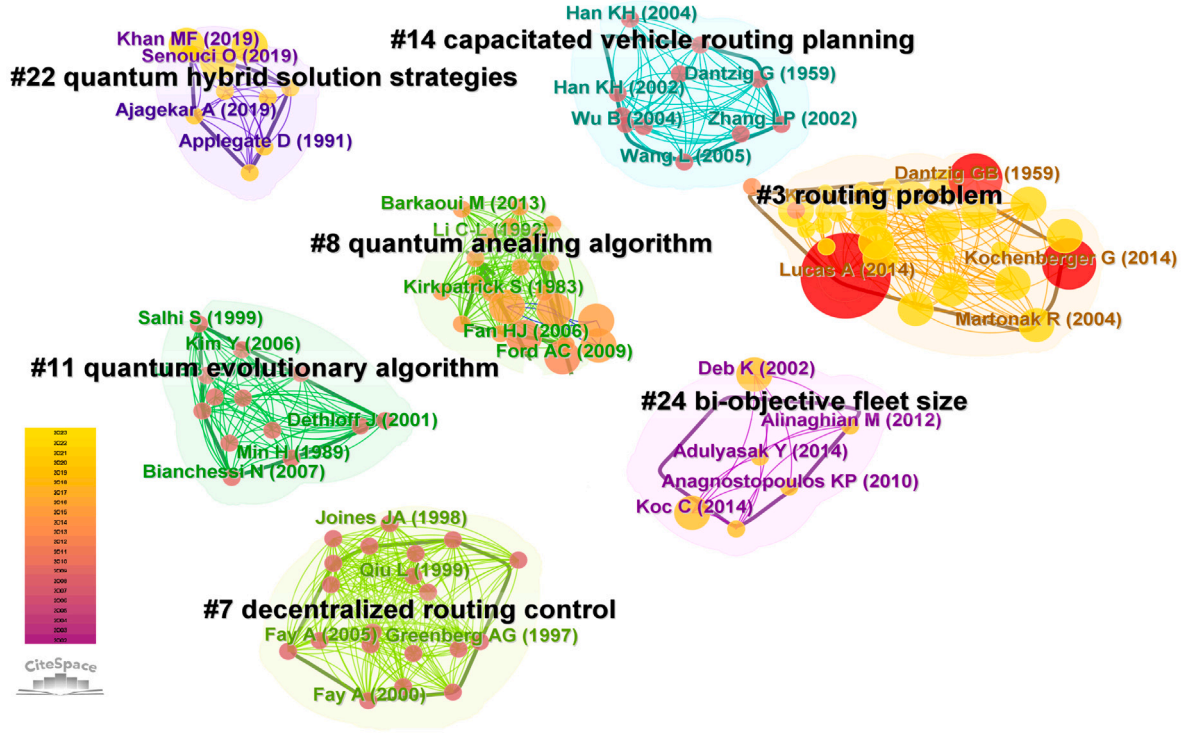
The investigation of geographical distribution analysis serves as a mechanism for delineating the participatory roles of diverse nations within a designated arena of research (Kaur et al., 2022). Therefore, this geographical distribution analysis plays a crucial role by highlighting the global spread of research activities and contributions in the QEITS field. This analysis identifies leading countries and regions in terms of publication output, revealing international collaborations and regional research strengths. Understanding the geographical distribution of research helps in recognizing influential research hubs, fostering international partnerships, and identifying less-explored areas that may benefit from increased focus and resources. This analytical procedure is predicated upon an assessment of publication frequencies originating from the top five countries within each discrete domain of the QEITS, as presented in Table 3. The presented table delineates the leading nations with their corresponding publication and share of publications within each category. Notably, China emerges as the foremost contributor across all QEITS domains, where the AVCDS domain receives a substantial 27.64% of its total publications solely from China. The USA has secured the second most significant position in all fields having % share 16.05, 34.57, 17.09, 21.06 in DVRN, AVCDS, STMF, and EVTS, respectively. India has generated the third most significant volume of scholarly papers in all domains. Japan has achieved the fourth-highest position in the field of DVRN (22.73%) and EVTS (34.09%) whereas Canada is the fourth-highest contributor in AVCDS and STMF. In the context of the fifth-highest contributor in the QEITS research domain, the UK lead in DVRN (20.45%), Italy in AVCDS (19.35%), South Korea in STMF (80.00%), and Germany in EVTS (45.00%) generated the largest proportion of publications.

5.3. Document co-citation network analysis

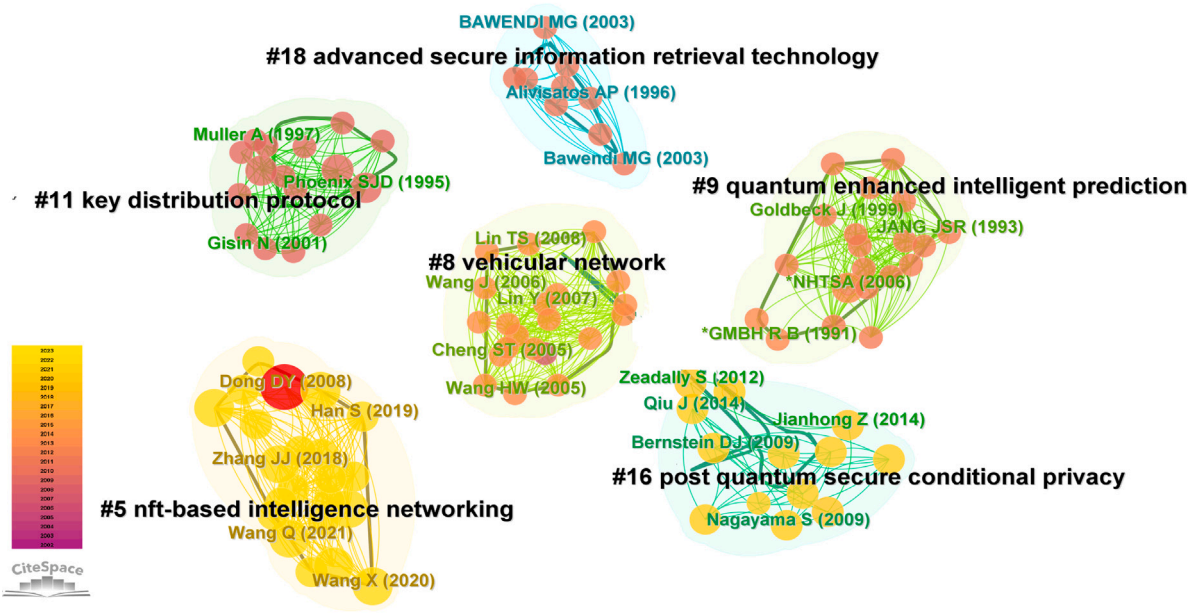
CiteSpace employs collaboration networks to depict the trajectories of QEITS research (Jia et al., 2019). Through DCoN Analysis, clusters of co-cited documents have been identified, unveiling the intellectual structure and principal research themes within each category of QEITS. This analysis elucidates the interrelationships among influential works, pinpointing emerging trends and focal points of research activity. Understanding these clusters enables researchers to navigate scholarly literature adeptly, discover interdisciplinary intersections, and strategically direct research efforts to foster advancements in QEITS knowledge. In the network diagram, highly referenced articles become nodes in a co-citation network. Interconnecting links indicate co-citation relationships. CiteSpace effectively presents the clusters using "Log-Likelihood Ratio" (LLR) algorithm by extracting indexed phrases from the key Refs. Sood and Chauhan (2023). The network's efficacy is gauged through its average silhouette value (S) for cluster consistency (ranging from −1 to 1) and modularity value (Q) for partition reliability (ranging from 0 to 1) (Jacob et al., 2023). In the network visualization, distinct color-shaded sections and nodes spotlight clusters within the pertinent QEITS domain. Cluster identification is denoted by numerical values with a hash symbol (#), and cluster tags, represented by titles followed by cluster IDs depict the associated research area (Su et al., 2019). In Table 4, an overview of distinct parameters for each domain is presented, including Cluster ID, Size, Silhouette Value, Mean Year, and the LLR label. The size metric identifies the volume of references citing each approach. The average publication year within a cluster offers insight into the prevalent publication period of the members. The LLR labels serve as indicators of the forefronts in the field, establishing distinctive identities for clusters that comprehensively encompass pivotal literature (Zaki et al., 2023). Concurrently, Figs. 9 and 10 visually depicts the network diagrams corresponding to each QEITS domain.

5.3.1. Intellectual structure of DVRN

The DVRN literature's DCoN encompasses 540 nodes, connected by 5483 links, yielding a network density of 0.0377. Network map



(a)

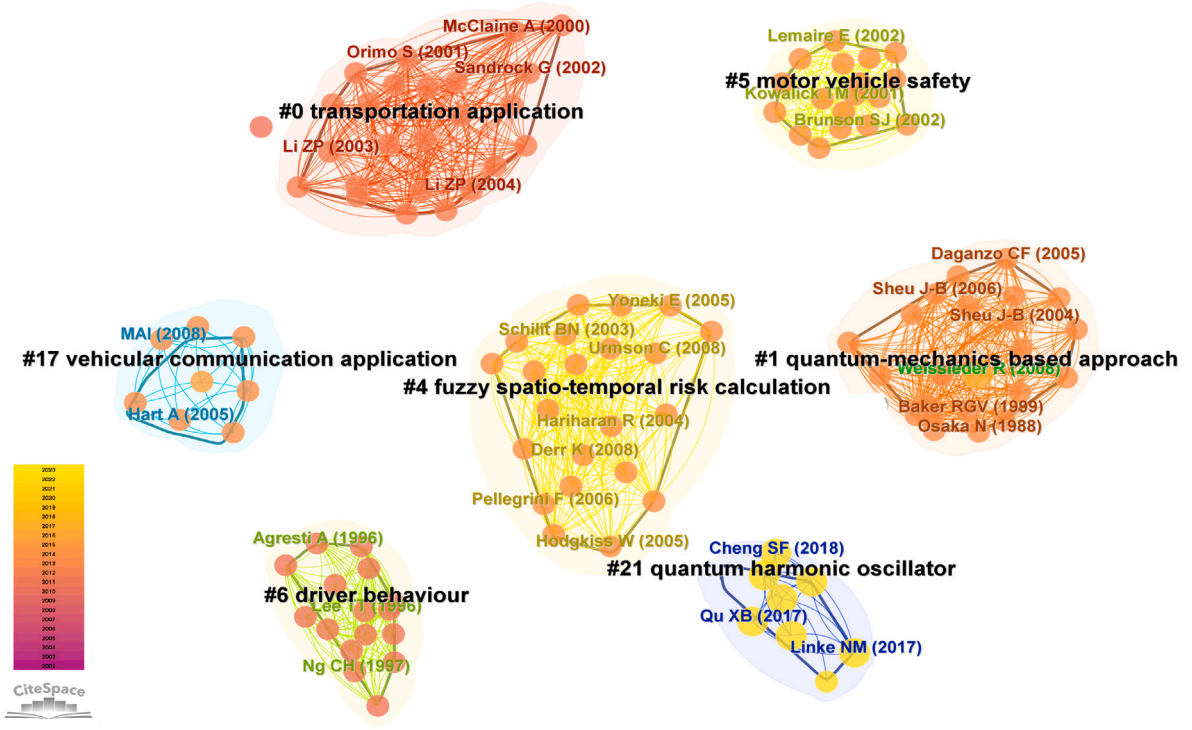


(b)

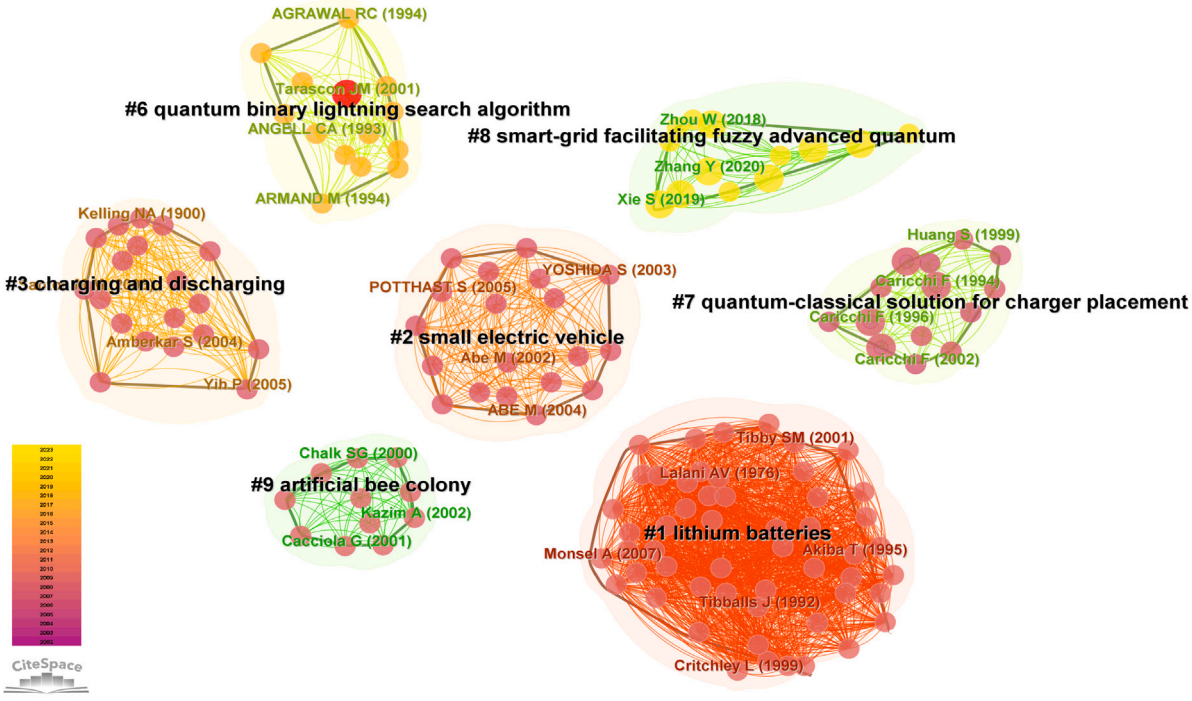
Fig. 9. Document co-citation network analysis: (a) DVRN (b) AVCDS.

values for S and Q stand at 0.9998 and 0.8359, respectively. Notably, the network comprises seven distinct clusters, indicative of various research frontiers. These clusters, along with corresponding metadata, are outlined in Table 4. The influential papers identified by CiteSpace that frame the clusters in this category are elaborated below to provide a comprehensive overview.

In the dynamic realm of solving routing problems, both classical and QC approaches have revolutionized the landscape with their distinct yet complementary methodologies. Classical methods, extensively reviewed by Elshaer and Awad (2020), have provided foundational metaheuristic algorithms like GA, "Particle Swarm Optimization" (PSO), and "Ant Colony Optimization" (ACO). These methods are well-established, practical, and widely used, offering detailed taxonomies and evaluations of their strengths and weaknesses. However, they do not address the emerging potential of QC. In contrast, Osaba et al. (2022) presented a systematic review of QC applications for routing problems. Further contributing to this field, Azad et al. (2022) specifically explored the



(a)



(b)

Fig. 10. Document co-citation network analysis: (a) STMF (b) EVTS.

use of Quantum Annealing and QAOA for solving the Vehicle Routing Problem (VRP), demonstrating its potential to significantly improve solution quality and computational efficiency. While QC holds promise for solving NP-hard problems more efficiently, its practical application is currently limited by hardware constraints. Nonetheless, Osaba et al. (2022) and Azad et al. (2022) highlighted the future potential of hybrid quantum–classical approaches and the need for technological advancements. Integrating insights from both classical and quantum perspectives can provide a more comprehensive approach to addressing the ongoing challenges in routing problems.

Table 4
Cluster information of each QEITS domain.

QEITS domain	Cluster ID	Size	Silhouette value	Mean year	LLR Label
DVRN	3	30	0.997	2014	Routing Problem
	7	21	1	1952	Decentralized Routing Control
	8	16	1	2003	Quantum Annealing Algorithm
	11	14	0.987	2004	Quantum Evolutionary Algorithm
	14	11	0.997	1998	Capacitated Vehicle Routing Planning
	22	7	1	2007	Quantum Enhanced Solution Strategies
	24	6	0.967	2011	Bi-objective Fleet Size
AVCDS	5	22	0.989	2009	NFT-based Intelligence Networking
	8	20	0.998	2004	Vehicular Network
	9	20	1	2001	Quantum Enhanced intelligent prediction
	11	16	0.967	1998	key distribution protocol
	16	9	1	2012	Post Quantum Secured Conditional Privacy
	18	9	0.997	2000	Advanced Secure Information Retrieval Technology
STMF	0	24	0.945	2002	Transportation Application
	1	23	1	1997	Quantum Mechanics-based Approach
	4	22	1	2005	Fuzzy Spatio-temporal Risk Calculation
	5	17	0.997	2001	Motor Vehicle Safety
	6	16	0.935	1993	Driver Behavior
	17	8	0.997	2007	Vehicular Communication Application
	21	8	1	2017	Quantum Harmonic Oscillator Model
EVTS	1	55	1	2003	Lithium Batteries
	2	19	0.987	1997	Small Electric Vehicle
	3	19	1	2006	Charging and Discharging
	4	18	1	2015	Quantum binary lightning search algorithm
	6	18	1	2017	Quantum Classical Solution for Charger Placement
	7	13	0.987	2007	Smart-grid Facilitating Fuzzy Advanced Quantum
	8	13	0.997	2017	Artificial Bee Colony

Decentralized routing control emerges as another pivotal innovation, revolutionizing adaptive and scalable solutions in dynamic environments. Claes et al. (2011) advocated for decentralized anticipatory vehicle routing using multiagent systems, where autonomous agents optimize routes based on real-time traffic conditions. Their approach highlights enhanced route efficiency and resource allocation compared to traditional centralized methods, laying the foundation for adaptive decision-making frameworks in logistics. Complementing this decentralized approach, Chen et al. (2024) introduced, a Decentralized Dynamic Congestion Avoid Routing Protocol for Quantum Networks (Q-DDCA). This protocol leverages QC principles to dynamically adjust routing paths, aiming to mitigate congestion and optimize network efficiency. However, practical implementation faces challenges such as quantum technology's error rates and scalability issues, necessitating further advancements in hardware capabilities.

Innovative adaptive strategies further augment routing efficiency in dynamic logistics scenarios. Barkaoui and Gendreau (2013) proposed an adaptive evolutionary approach for real-time vehicle routing and dispatching, integrating evolutionary algorithms with adaptive mechanisms to dynamically adjust routes based on real-time data. This method enhances routing efficiency and service quality, addressing the complexities of real-time logistics operations. Concurrently, Hu and Wu (2009) pioneer a "Quantum Evolutionary Algorithm" tailored for the VRP with simultaneous delivery and pickup tasks, leveraging quantum mechanics to optimize routing efficiency under complex logistics scenarios. Similarly, Feld et al. (2019) introduced a hybrid solution method for the Capacitated Vehicle Routing Problem, integrating quantum annealing with classical algorithms to optimize vehicle routing under capacity constraints. These advancements underscore the transformative potential of quantum-inspired algorithms in revolutionizing logistics operations, enhancing efficiency, and improving solution quality across diverse routing challenges. Moreover, Ajagekar et al. (2020) explored hybrid models that combine deterministic algorithms with QC techniques, demonstrating superior computational efficiency in solving large-scale mixed-integer programming problems and vehicle routing instances. Additionally, Alinaghian and Zamani (2019) proposed a bi-objective model for the Green Inventory Routing Problem (IRP) with a heterogeneous fleet, leveraging a quantum evolutionary algorithm to minimize emissions and fleet size while optimizing routing and inventory costs. These studies collectively highlight the forefront of innovation in routing optimization, showcasing how quantum and hybrid methodologies are reshaping logistics and transportation sectors with unprecedented efficiency and scalability.

Classical and QC offer distinct yet complementary solutions to routing problems, each with unique advantages. Classical methods, such as GA, PSO, and ACO, are well-established and practical. In contrast, quantum approaches, like Quantum Annealing and the QAOA, promise improved solution quality and efficiency but are currently limited by hardware constraints. Decentralized routing control, exemplified by multiagent systems and quantum congestion avoidance protocols, enhances adaptability and efficiency in dynamic environments. Adaptive strategies, including hybrid models that integrate classical and quantum techniques, further optimize routing in real-time logistics. Combining these insights can revolutionize routing problem solutions, providing unprecedented efficiency and scalability in the logistics and transportation sectors.

5.3.2. *Intellectual structure of AVCDS*

The DCoN in the AVCDS literature comprises 530 nodes connected by 3058 links, resulting in 0.0218 network density. The S and Q values of the network are 0.9962 and 0.9372, respectively. Notably, the network contains six distinct clusters, each representing different research areas. Additional information about these clusters is presented in Table 4. The notable papers identified by CiteSpace that define the clusters in this category are explained below to give a comprehensive overview.

In this category, Ren et al. (2022) presented a highly cited paper introducing an innovative ("Non-fungible Token") NFT-based Distributed Intelligence Networking Scheme (NDIN) for connected and autonomous vehicles (CAVs) in the Internet of Vehicles (IoV). By using non-fungible tokens to tokenize and describe intelligence, NDIN allows CAVs to share and utilize distributed knowledge efficiently. The scheme formulates the problem as a discrete Markov decision process and uses a Quantum-inspired Reinforcement Learning (QRL) algorithm for optimized decision-making, resulting in faster learning and convergence. Simulations show NDIN's superior performance in efficiency, scalability, and learning speed, while also addressing challenges in intelligence

description, security, and privacy. The study highlights NDIN's potential to enhance CAV capabilities in complex environments and suggests future research on energy-efficient collaborative learning strategies. As communication is the most important part of the vehicular network, scientists are working to build a proper network. Furthermore, addressing the critical aspect of communication within vehicular networks, Chang et al. (2010) focused upon "Quantum-enhanced Intelligent Prediction". This study proposes a fuzzy fault-tolerant approach to integrate heterogeneous sensed data for more accurate collision warnings in pre-crash systems. It considers drowsy driving risks by factoring in fatigue level, carbon monoxide, and breath alcohol concentration, adjusting algorithms for enhanced reaction time. Vision-based lane marking detection and collision warnings, supported by a Global Positioning System and inter-vehicle communication, are included. The system records event data for accident reconstruction and accelerates data fusion using a Quantum-tuned Back-propagation Neural Network and Adaptive Network-based Fuzzy Inference System. Testing demonstrates improved detection time and reduced accident rates compared to AWS-Mobileye and ACWS-Delphi systems.

Moreover, in the realm of security within vehicular networks, Deng and Long (2004) shed light on vehicular networks as well as on security. The paper presented a protocol for Quantum Secure Direct Communication (QSDC) using single photons, designed for direct transmission of secret messages without the need to establish a shared secret key beforehand. This method involves encoding messages using single photons prepared in one of four states, serving as a one-time pad. The proposed scheme ensures unconditional security and is feasible with current technology. Unlike Quantum Key Distribution (QKD), which focuses on creating a common secret key, QSDC directly transmits messages, which can be crucial in urgent scenarios. The protocol's security is analyzed and shown to be robust against eavesdropping by employing a method similar to the BB84 QKD scheme. Single photons are first sent from Bob to Alice, who then checks for eavesdropping before encoding and returning the photons. This process guarantees that any interception can be detected before the message is revealed. The protocol's practicality and security make it a promising solution for secure communication in quantum networks. Meanwhile, Chang et al. (2008) explored about "Advanced Secure Information Retrieval Technology". The research investigates colloidal semiconductor Quantum Dots (QDs) synthesized via wet chemistry for secure information retrieval. A key merit is their application in multiplexed spectral coding, leveraging QDs' optical properties like narrow emission bandwidth and photostability. Methodologically, rigorous experiments validate QD-based "infoinks" on diverse substrates, confirming their feasibility in real-world security applications. However, challenges include optimizing QD synthesis for specific spectral needs and scaling production. Despite these, the study underscores QDs' potential to enhance security protocols through innovative information encryption and retrieval solutions.

In the current research landscape, quantum cryptography and post-quantum cryptography are pivotal fields driving advancements in vehicular network security. Bourennane et al. (2000) explored long wavelength "plug and play" quantum cryptography systems and their single-photon detector performance. The implementation of quantum cryptography protocols, focusing on error correction and privacy amplification to achieve a secure key is discussed. Through software simulations, the complete protocol's performance in terms of the final secure key creation rate is illustrated. By combining experimental and simulation results, the study shows how critical parameters such as the final key rate and transmission distance impact system capacity. Despite some challenges, the feasibility of 1550 nm quantum cryptographic systems is demonstrated. With improvements in detector performance and key rates, these systems could support practical telecom applications over long distances. Additionally, Djordjevic (2020) explored the integration of disconnected terrestrial Quantum Communication Networks (QCNs) through Low Earth Orbit (LEO) satellite systems, presenting a novel approach to enhance QKD and secure information exchange. The use of cluster state protocols forms the backbone of this heterogeneous satellite–terrestrial QCN, offering several merits. Cluster states enable robust quantum communication across various applications, including quantum teleportation, distributed QC, and quantum sensing networks. Moreover, the proposed architecture promises unprecedented security levels for future wireless networks, the Internet of Things (IoT), and autonomous vehicles, leveraging physical principles to ensure data integrity. However, challenges such as channel impairments and technological integration complexities are notable demerits. Methodologically, the paper integrates theoretical frameworks with practical implementations, detailing the development of the QCN using cluster states and LEO satellite networks. Findings indicate significant advancements in extending QKD distances and enhancing network resilience against channel turbulences. This research lays the foundation for scalable and secure quantum communication infrastructures, essential for next-generation cyber security systems and beyond.

Various models in vehicular and quantum networks offer complementary solutions with unique advantages. An NFT-based Distributed Intelligence Networking Scheme (NDIN) enhances decision-making and learning speed for Connected and Autonomous Vehicles (CAVs) in the IoV. Quantum-enhanced intelligent prediction systems, integrating fuzzy logic and quantum-tuned neural networks, improve collision warnings and driver safety. For secure communication, QSDC protocols provide robust security against eavesdropping, while QDs enable advanced encryption for secure information retrieval. Quantum cryptographic systems and LEO satellite networks extend secure key distribution distances and enhance network resilience. Together, these models significantly advance efficiency, security, and scalability in next-generation IoT and autonomous vehicle applications, ensuring robust and secure communication and improved vehicular performance.

5.3.3. Intellectual structure of STMF

The DCoN described in the STMF literature consists of 473 nodes linked by 2375 connections, yielding 0.0213 network density. Additionally, the network has S and Q values of 1 and 0.9355, respectively. Notably, this network comprises seven distinct clusters, each corresponding to various research domains, and further details about these clusters are in Table 4. To provide a comprehensive overview of this category, the top influential papers are described below.

The foremost paper by Chintalacheruvu et al. (2012) introduced an advanced video-based vehicle detection system utilizing the Harris-Stephen corner detector algorithm for ITS. Methodologically, the system processes live video feeds at 29.95 frames per second using a Hauppauge Live USB frame grabber device, with frames sized at 320 × 240 pixels. It employs non-maxima suppression and thresholding to identify interest points, tracking vehicle centroids to determine counts and speeds in roadway segments and lanes. The system shows robust performance in vehicle counting and speed measurement, comparable to commercial systems, though challenges like camera field of view skew affect accuracy in distant lane counts. Findings underscore its effectiveness for real-time ITS applications, particularly in warning systems for congestion and speed reductions at work zones and events. Future research aims to optimize camera placement and implement shadow elimination algorithms for enhanced accuracy in varying traffic conditions. Building on the theme of ITS and its application to driver behavior, Lum and Wong (2003) investigated the impact of Red Light Cameras (RLCs) on driver behavior at signalized intersections through a before-and-after analysis. Using data loggers and loop sensors, traffic parameters and driver responses to amber signals were recorded. Logistic regression models were employed to analyze various factors such as distance from the stop-line, vehicle speed, and the presence of RLCs. The findings indicate that RLCs significantly increase the likelihood of drivers stopping at camera-designated approaches, particularly evident at T-intersections where

the distance from the stop-line interacts with RLC installation. Conversely, non-camera approaches showed minimal effects from RLCs. Model validation demonstrated high predictive accuracy, underscoring the efficacy of logistic regression in comprehending and forecasting driver behavior in complex intersection scenarios. Expanding on the analysis of driver behavior and introducing a novel theoretical approach, Sheu (2008) proposed a quantum mechanics-inspired model to simulate driver behavior near lane-blocking incidents, aiming to enhance road safety. The study breaks down the driver decision-making process into three phases: initial stimulus perception, glancing-around car-following, and incident-induced driving behavior. The model, validated with real-world data from a Taiwanese freeway, effectively captures incident-induced driving dynamics. Results show promising alignment between simulated and observed traffic behaviors, highlighting potential improvements in incident management and driver safety strategies. Future research should focus on refining the model's assumptions and testing it in diverse traffic scenarios to further enhance its applicability and accuracy.

Furthermore, to minimize accidents and enhance driver safety, Shannon et al. (2018) investigated the complex interplay between vehicular crash factors and expected trauma compensation payouts using data from the National Highway Traffic Safety Administration's Crash Injury Research database and Ireland's Book of Quantum guidelines. Through multiple linear regression analysis and comparison with ordered and multinomial logit models, the study identifies key variables significantly influencing compensation costs, such as relative speed at impact, availability of airbags, vehicle type, lighting conditions, and crash category. The findings underscore the impact of Advanced Driver Assistance Systems on mitigating common crash factors, suggesting potential shifts in insurance risk assessments and claims reserving strategies. Further enhancing the focus on technological advancements for road safety, Costa et al. (2008) explored the potential of WiMAX technology for enhancing road safety through vehicular communication applications. Given the significant social and economic impacts of road accidents, effective communication among vehicles and with roadside infrastructure is crucial. WiMAX, known for its Quality-of-Service (QoS) capabilities and traffic differentiation features, presents an attractive option despite being originally developed for broadband access. The study reviews current research in vehicular communications, proposes Vehicle-to-Infrastructure (V2I/I2V) services achievable with WiMAX, and presents initial experimental results. These findings suggest that WiMAX could effectively support critical road safety and traffic management services, paving the way for further investigation and deployment in this important domain. Additionally, Derr and Manic (2008) introduced the Dynamic Spatio-Temporal for Significant Place Extraction (DSTiPE) algorithm for extracting significant places from wireless spatio-temporal data, crucial for applications in factory automation, autonomous vehicles, and urban mobility. Using a distance-based clustering approach, DSTiPE identifies places where individuals or objects spend significant time, aiding in urban dynamics understanding, emergency planning, and process optimization. A fuzzy inference system then evaluates risk based on proximity and duration, validated using synthetic data. Future work aims to refine thresholds and explore multi-resolution approaches, enhancing their utility across domains. This integration of spatio-temporal data with fuzzy risk assessment offers innovative solutions for smart cities and industrial optimization.

In the realm of traffic simulation and building on previous models, Hu et al. (2022) introduced a novel method, named Quantum Harmonic Oscillator Model for Fine-grained Expressway Traffic Volume Simulation (FGTVS-QHO), addressing challenges in accurately modeling individualized vehicle behaviors crucial for ITS. By leveraging the quantum harmonic oscillator model's wave function, FGTVS-QHO probabilistically describes vehicle position dynamics and applies optimization strategies for numerical solutions. Validated on the Nanjing-Shanghai Expressway, FGTVS-QHO outperforms Long Short-Term Memory and AutoRegressive Integrated Moving Average methods with higher accuracy, lower time complexity, and reduced parameter complexity. Its scalability and adaptability for fine-time granularity simulations make it promising for applications in traffic control, management, autonomous driving, and unmanned driving. Future research aims to integrate abnormal conditions and enhance analytical solutions to further refine FGTVS-QHO's effectiveness in traffic volume simulations.

Various models in ITS and vehicular communication offer unique solutions. An advanced video-based vehicle detection system uses the Harris-Stephen corner detector for real-time counting and speed measurement, despite challenges with the camera field of view. In contrast, a quantum-inspired driver behavior model enhances road safety by simulating incident-induced dynamics effectively. RLCs significantly influence driver behavior at intersections, improving compliance at camera-designated approaches. Multiple linear regression analyzes crash factors and compensation payouts, highlighting ADAS's role in mitigating crash risks. WiMAX technology supports critical road safety and traffic management with its QoS capabilities. The DSTiPE algorithm identifies significant places from spatio-temporal data, optimizing urban dynamics. Lastly, the FGTVS-QHO excels in traffic volume simulations with higher accuracy and scalability. These models collectively advance ITS, each offering distinct benefits in enhancing road safety, traffic management, and vehicle detection technologies.

5.3.4. Intellectual structure of EVTS

In the EVTS literature, the DCoN is composed of 528 nodes connected by 13,316 links, resulting in 0.0957 network density. Furthermore, the network exhibits S and Q values of 1 and 0.3596, respectively. Importantly, this network encompasses seven separate clusters, each associated with different research areas. Furthermore, Table 4 provides additional information about these clusters and to offer a thorough overview of this category, the most influential papers are detailed below.

EVs are rapidly advancing, with significant technological innovations and increased adoption worldwide. As a key component in sustainable transportation, they are transforming the automotive industry and contributing to global efforts to reduce carbon emissions. For the electric vehicle development, Santos et al. (2006) introduced a fault-tolerant modular control architecture designed for an electrical vehicle (VEIL) equipped with x-by-wire subsystems, including steer-by-wire, brake-by-wire, and accelerate-by-wire functionalities. VEIL aims to advance urban mobility with a license-free electric vehicle that integrates automated control systems for braking, steering, and acceleration. The architecture leverages commercial off-the-shelf components and the FTT-CAN protocol for synchronized communication and task scheduling, essential for safety-critical applications in modern automotive design. The transition from mechanical to electronic systems not only reduces vehicle weight and complexity but also enhances energy efficiency and operational reliability, aligning with trends towards sustainable urban transportation solutions. Researchers aim to develop sustainable batteries to meet modern power demands, while addressing environmental concerns and material limitations (Armand and Tarascon, 2008). In the context of lithium batteries, Lee et al. (2007) described that cation (Mg and Al) and anion (F) substituted spinel were synthesized via co-precipitation using metal oxide precursors. X-ray diffraction confirmed the spinel structure (Fd3m space group), indicating successful dopant incorporation without distortion. Scanning electron microscopy revealed spherical morphology with varying particle sizes. At 55 °C, all substituted spinels showed improved cycling stability. This study highlights the effectiveness of cation and anion substitution in enhancing spinel cathode stability for lithium-ion batteries under elevated temperatures.

Charging and discharging of EVs are critical processes that determine their usability and efficiency. Advances in battery technology and infrastructure are pivotal in enhancing the convenience and range of EVs. Lu et al. (2013) explored the magnetic properties of lithium

peroxide in nonaqueous lithium-oxygen batteries, crucial for advancing battery technology with high theoretical energy density for EVs. Key findings include the unexpected magnetic moment in lithium peroxide, confirmed through dc-magnetization, Density Functional Theory (DFT) calculations, and EPR spectroscopy. Methodologically, the study uses an ether-based electrolyte (LiCF3SO3-1NM3) and Super P carbon cathode in a coin cell setup, supported by X-ray Photoelectron Spectroscopy (XPS), X-ray Absorption Near-edge Structure (XANES), EPR, and Superconducting Quantum Interference Device (SQUID) analyses. While shedding light on lithium peroxide behavior, potential influences from electrolyte decomposition on magnetic properties require further investigation. Overall, the research provides insights into enhancing lithium-oxygen battery performance and stability, paving the way for future energy storage advancements. Additionally for utilizing quantum advantage in this area, Aljanad et al. (2018) utilized the Quantum Binary Lightning Search Algorithm (QBLSA). Vehicle-to-Grid (V2G) technology in Plug-in Hybrid EVs (PHEVs) has emerged as a significant resource for backup power in distribution networks during discharging. The placement of Charging Stations (CSs) in these networks is critical to optimizing V2G utilization and enhancing network performance during peak demand. The paper proposed a methodology using the QBLSA, a novel heuristic optimization technique, to determine optimal CS placement. A multi-objective function minimizes line loading, voltage deviation, and circuit power loss, leveraging V2G capabilities. Comparative analysis with other heuristic methods demonstrates QBLSA's reduced likelihood of premature convergence. The study underscores PHEV's role in grid support under peak load conditions and emphasizes the importance of strategic CS placement in optimizing network performance and preparing for future peak demands. The placement of EVCSs plays an important role in supporting the increasing adoption of EVs while meeting the requirements of EV owners, energy grids, and operational entities. Traditional computational methods face scalability issues when solving the complex combinatorial optimization problem of EVCS placement. To address these challenges, Rao and Sodhi (2023) introduced a novel quantum–classical hybrid solution. By utilizing QC, specifically quantum annealing and gate-based quantum circuits, the proposed approach achieves a significant computational speedup, surpassing traditional methods by more than 500%. Despite inherent challenges such as fixed initial location assumptions and complexities in demand modeling, our scalable approach effectively identifies optimal EVCS placements. All software artifacts necessary for validation are openly shared, facilitating future enhancements that may incorporate additional parameters like charging times and road network distances.

Electric vehicle energy management involves balancing electrical power consumption from the vehicle's battery, thermal power requirements for heating and cooling systems, and managing overall energy loads to optimize efficiency and range. Integrating these aspects ensures effective utilization of electric vehicle resources while enhancing user comfort and sustainability. In this context, Cai et al. (2019) introduced a method for programming an Energy Hub (EH) that manages electrical power, thermal power, and cooling loads, integrating continuous and on/off controllable utilities. The model comprehensively addresses features such as power losses, amortization costs of energy storage systems, and operational constraints of mixed energy units. Applied across two typical summer and winter days under various scenarios, the model evaluates the impact of energy storage systems, power supply to the grid, intelligent EV charge/discharge strategies, and controllable utility management. Numerical results demonstrate significant benefits for users and the power network through optimized EH utilization and load management. Specifically, EV integration is shown to smooth load profiles during Demand Response Programs, enhancing overall system efficiency. Additionally, Chakraborty et al. (2013) presented an intelligent economic operation strategy for a smart grid using an advanced quantum-inspired evolutionary method. The strategy integrates wind and photovoltaic generation with thermal generators to meet consumer demand efficiently and mitigate global warming effects. PHEVs introduced uncertainties managed through fuzzy logic-based formulations. The proposed Intelligent Quantum-inspired Evolutionary Algorithm (IQEA) optimizes the scheduling and dispatching of thermal units, wind farms, photovoltaic arrays, and PHEVs using sophisticated operators like rotation and differential operators. Simulation on a hypothetical power system demonstrates IQEA's effectiveness in reducing production costs and minimizing emissions. Future research will focus on scaling the method to larger smart-grid networks, leveraging parallel computing to handle increased computational complexity.

EVs are advancing sustainability with innovations across various domains. A modular control architecture enhances EV's reliability and efficiency through electronic systems. Improvements in lithium-ion battery stability and high-energy-density applications are facilitated by spinel cathode substitutions and exploration of lithium peroxide battery magnetic properties. Optimization of V2G technology strategically places Charging Stations (CSs) for grid performance, while QC accelerates efficient EVCS placement. An EH management model integrates EV charge strategies to optimize power and thermal loads effectively, and IQEA optimizes smart grid operations, minimizing emissions and enhancing renewable energy integration. These innovations underscore EV's pivotal role in sustainable mobility, driving advancements in battery technology, charging infrastructure, and energy management systems for future urban transport solutions.

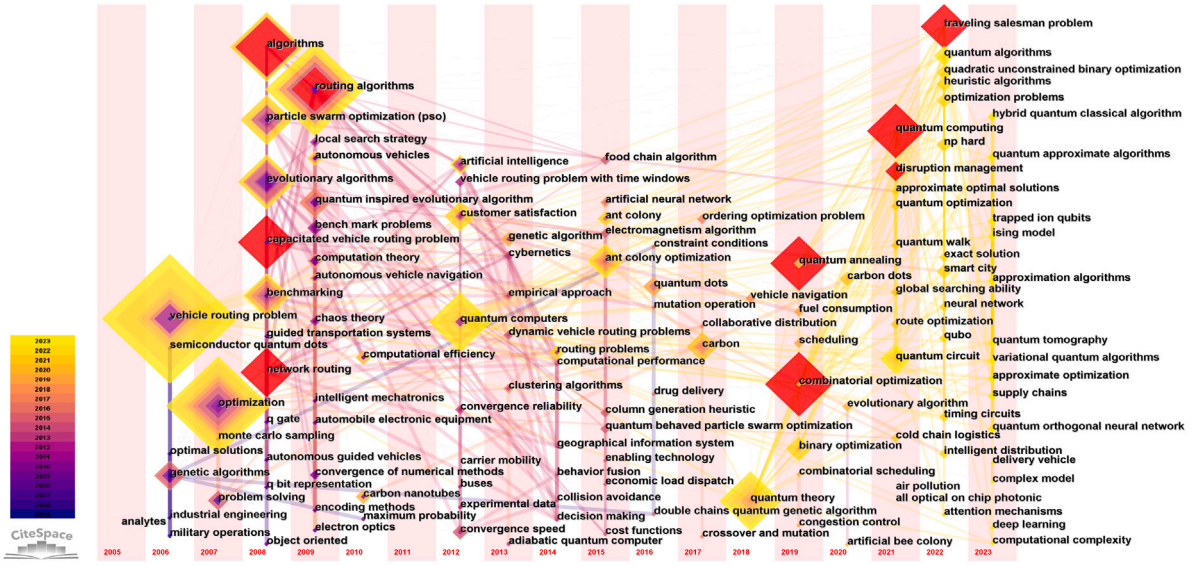
5.4. Keyword co-occurrence network analysis

KCoN analysis delivers significant insights into emerging areas in a concise manner (Kumar et al., 2023). KCoN analysis reveals central themes and emerging trends within QEITS. The generated KCoNs offer a comprehensive overview by illustrating correlations between keywords. Furthermore, by identifying frequently co-occurring keywords and those experiencing bursts of interest, this analysis aids in understanding current research focuses, guiding strategic decision-making, and fostering interdisciplinary collaborations. Employing the CiteSpace tool, KCoN analysis is executed for each delineated category of QEITS. The KCoN visualizations in Figs. 11 and 12, portray the KCoN of QEITS across different categories.

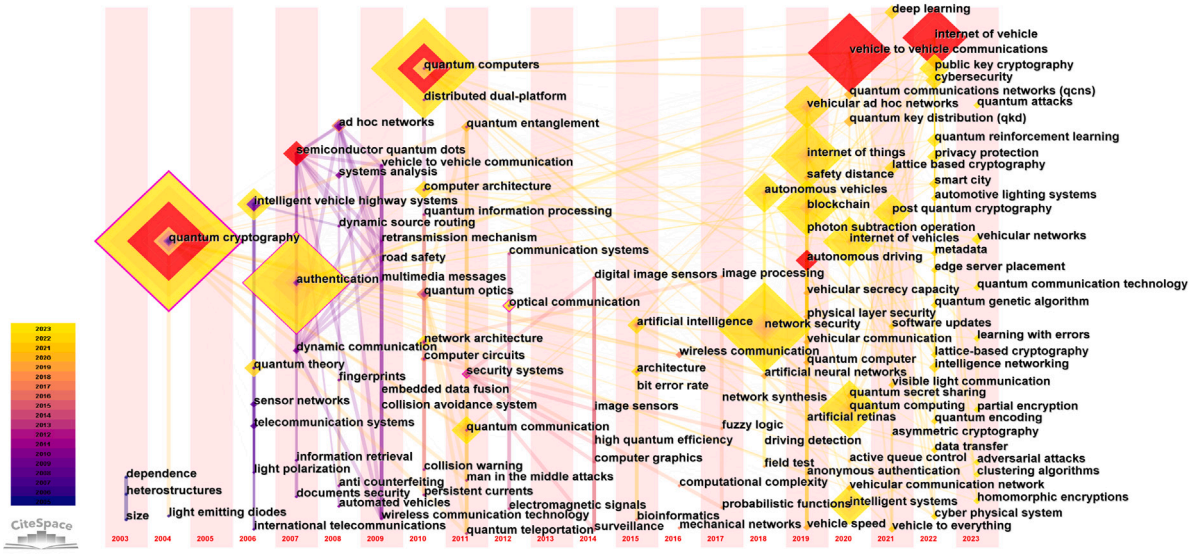
In these visual representations, individual nodes symbolize keywords, with their dimensions indicative of the frequency of their appearance. Node size in KCoN corresponds to frequency, with larger nodes indicating active research domains. In KCoN, the degree signifies keyword associations in the dataset, revealing connectivity and prominence. The external pink layer of the nodes shows the centrality of the keyword, darker the ring more influential the keyword. This metric underscores potential importance in the research arena. Table 5 showcases highly frequent keywords, accompanied by degree and centrality values. Moreover, AI serves as a ubiquitous term in all network diagrams, symbolizing its fundamental role in rendering transportation systems intelligent. Another important node indicator, known as "burst", measures the intensity of usage surges for terms that displayed notable fluctuations in usage frequency within the designated time frame (Sood and Pooja, 2023). This metric represents the abrupt surge in citations during a specific period within that domain. Therefore, Fig. 13 represents the burst keywords with the strongest burst value and years. In Figs. 13(a), 13(b), 13(c), and 13(d), it is observed that, particle swarm optimization (Rehman et al., 2019), smart city, post-quantum cryptography in-vehicle communication, drug delivery, smart parking, energy management system for EVs, charging and discharging facilities, lithium-ion batteries, AI, solar energy are the most impactful subtopics in this field of knowledge.

6. Discussion

The current study utilizes a scientometric literature analysis methodology to delve into the prior research inquiries surrounding the QEITS research field. This analysis is particularly concentrated on



(a)



(b)

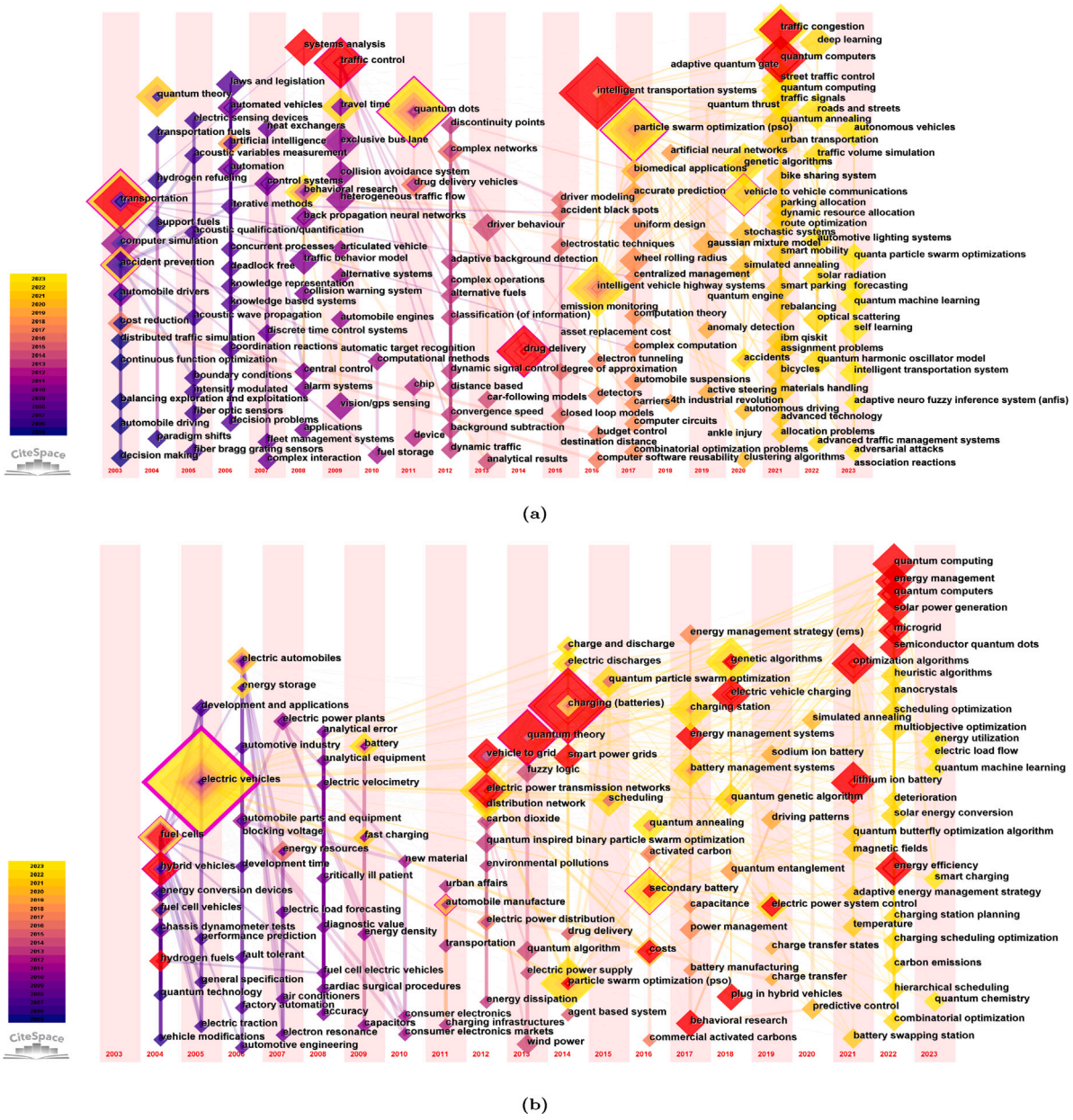
Fig. 11. Keyword co-occurrence network analysis: (a) DVRN (b) AVCDS.

Table 5
Top five keywords in each QEITS domain.

Frequency	Degree	Centrality	Keyword	Frequency	Degree	Centrality	Keyword
DVRN				**AVCDS**			
26	104	0.09	Vehicle Routing Problem	38	139	0.32	Quantum Cryptography
20	101	0.09	Optimization	25	96	0.10	Network Security
19	94	0.07	Routing Algorithms	17	82	0.02	Internet of Vehicle
12	64	0.04	Quantum computers	16	74	0.08	Block Chain
11	63	0.04	Network Routing	11	57	0.05	Quantum Communication
STMF				**EVTS**			
19	63	0.09	Intelligent Systems	62	257	0.87	Electric Vehicles
12	61	0.30	Transportation	32	120	0.15	charging(Batteries)
15	47	0.30	Quantum Dots	20	82	0.08	Optimization
8	41	0.03	Traffic Congestion	7	81	0.16	Hybrid Vehicles
7	37	0.02	Automated Vehicles	7	32	0.02	Vehicle to Grid

two comprehensive databases, namely Scopus and WoS, spanning from their inception. The investigation is systematically organized into four well-defined categories: DVRN, AVCDS, STMF, and EVTS.

(i) To address the initial research query about publication trends, the authors have conducted a comprehensive analysis of publication patterns within the realm of QEITS. The examination of



(a)

(b)

Fig. 12. Keyword co-occurrence network analysis: (a) STMF (b) EVTS.

literature distribution across distinct categories reveals a noteworthy surge in research over the past decade, signifying a nascent but highly promising knowledge domain. A meticulous assessment of publication trends in each category underscores the dominance of AVCDS with the highest publication count (313). The exploration and utilization of diverse quantum technology techniques present a notable avenue for substantial advancements and paradigm-shifting changes within the transportation field.

(ii) In addressing the second research query concerning countries with substantial contributions. A clear pattern emerges where China, the USA, and India take the forefront as predominant contributors in every QEITS domain. Additionally, notable engagement is observed from Japan, Canada, the UK, Italy, South Korea, and Germany.

(iii) In addressing research question three, DCoN analysis elucidates the intellectual structure within each QEITS domain. Notably, the analysis reveals that in the DVRN, routing problems and quantum-enhanced solution strategies have emerged as the most recent clusters in 2022 and 2023, signifying their significance as active research topics and recent trends. Each cluster represents a distinct subtopic within its respective domain. Moreover, within the QEITS domains, specific subtopics have emerged as particularly active and noteworthy. These include NFT-based intelligent clustering and post-quantum secured conditional privacy in the AVCDS domain, research into vehicular communication applications and the development of the quantum harmonic oscillator model in the STMF domain, and the exploration of quantum–classical solutions for charger placement and the utilization of the quantum artificial bee colony approach in the EVTS domain. Additionally, KCoN analysis identifies the research frontiers such as vehicle routing problems, optimization, evolutionary algorithms, quantum cryptography, transportation, QDs, and electric automobiles in the QEITS domain. It also recognizes that semiconductor quantum dot and carbon quantum dot qubits, as well as quantum annealing phenomena, are predominantly utilized and AI is an influential research topic.

Keywords	Year	Strength	Begin	End	2005 - 2023
genetic algorithms	2006	1.84	**2006**	2016	
quantum inspired evolutionary algorithm	2009	2.04	**2009**	2017	
routing algorithms	2009	2.44	**2012**	2014	
capacitated vehicle routing problem	2008	2.69	**2013**	2020	
quantum dots	2016	1.98	**2016**	2019	
particle swarm optimization	2019	1.84	**2019**	2023	
traveling salesman problem	2022	3.67	**2022**	2023	
quantum annealing	2019	3.08	**2022**	2023	
smart city	2022	1.52	**2022**	2023	

(a)

Keywords	Year	Strength	Begin	End	2003 - 2023
intelligent vehicle highway systems	2006	2.04	**2006**	2008	
semiconductor quantum dots	2007	4.38	**2007**	2009	
vehicular ad hoc networks	2008	1.96	**2008**	2023	
quantum optics	2010	2.26	**2010**	2016	
autonomous driving	2019	2.76	**2019**	2021	
vehicle to vehicle communications	2020	5.06	**2020**	2023	
quantum cryptography	2004	3.43	**2020**	2021	
quantum key distribution	2020	1.88	**2020**	2020	
internet of things	2022	1.58	**2022**	2023	
internet of vehicles	2020	3.97	**2022**	2023	

(b)

Keywords	Year	Strength	Begin	End	2003 - 2023
computer simulation	2003	1.7	**2003**	2010	
accident prevention	2003	1.67	**2003**	2006	
semiconductor quantum dots	2008	3.31	**2008**	2009	
traffic control	2009	2.59	**2009**	2018	
exclusive bus lane	2009	2.44	**2009**	2010	
drug delivery	2014	3.98	**2014**	2018	
intelligent vehicle highway systems	2016	2.05	**2016**	2021	
intelligent transportation systems	2016	2.65	**2019**	2023	
authentication	2020	1.99	**2020**	2023	
smart parking	2021	3.22	**2021**	2023	

(c)

Keywords	Year	Strength	Begin	End	2003 - 2023
fuel cells	2004	3.23	**2004**	2008	
electric utility	2007	2.11	**2007**	2013	
electric power transmission networks	2012	2.69	**2012**	2018	
smart power grids	2014	2.5	**2014**	2015	
secondary battery	2016	1.66	**2016**	2016	
energy management systems	2017	1.64	**2017**	2023	
plug in hybrid vehicles	2018	1.85	**2018**	2021	
electric vehicle charging	2018	1.74	**2018**	2023	
lithium-ion battery	2021	3.72	**2021**	2023	
solar energy	2022	2.72	**2022**	2023	

(d)

Fig. 13. Top 10 keywords with the strongest citation bursts: (a) DVRN (b) AVCDS (c) STMF (d) EVTS.

(iv) In addressing the fourth research question, the subsequent subsection delves into an exploration of the primary key challenges and outlines potential future research directions.

7. Research challenges and future research directions

The current research employs a scientometric approach to comprehensively assess scholarly literature, aiming to provide insights into the current landscape of QEITS. Through this analysis, several prominent challenges as well as solutions within ITS have been identified. The conducted analysis underscores the significant potential of quantum technology within the realm of transportation. Quantum technology has the potential to revolutionize ITS by optimizing traffic management, enhancing data security, accelerating machine learning processes, optimizing energy usage, and predicting traffic patterns. It can also contribute to lightweight vehicle materials, environmental impact assessment, advanced sensors, logistics optimization, and autonomous vehicle development. Quantum technology's integration promises significant advancements in safety, efficiency, and sustainability across transportation networks. Furthermore, the literature analysis uncovers the following research areas and gaps:

(i) The Vehicle Routing Problem is indeed a significant challenge in logistics optimization. The Vehicle Routing Problem challenges researchers to optimize vehicle allocation for efficient logistics. To enhance this process, future research aims to develop advanced algorithms such as QGA, quantum-inspired evolutionary algorithms, QAOA, PSO, and quantum annealing. Leveraging the computational power and adaptive capabilities of these techniques can significantly improve route planning and resource allocation within transportation networks. Key studies, including those by Azad et al. (2022) and Sinno et al. (2023), provide valuable insights for integrating quantum and evolutionary principles to enhance vehicle routing optimization.

(ii) As vehicular networks become increasingly interconnected, they face heightened risks from cyber threats that compromise data integrity and user privacy. To enhance vehicular network security comprehensively, a multifaceted strategy is essential. Incorporating advanced encryption protocols, notably post-quantum cryptography such as "lattice-based or hash-based cryptography" (Al-Mekhlafi et al., 2024; Sutradhar, 2023), forms a robust foundation. Augmenting these measures with sophisticated authentication mechanisms like digital signatures and biometric authentication enhances security further. Additionally, leveraging machine learning algorithms for anomaly detection and integrating blockchain technology for secure data sharing reinforces the network's resilience against evolving cyber threats (Shim, 2021).

(iii) Accurate prediction of real-time traffic patterns is essential for effective traffic management, yet traditional models often fall short due to the complexity of traffic data. Developing quantum machine learning models is crucial for accurately predicting real-time traffic patterns. Traditional models often struggle with the complexity of traffic data, leading to less precise predictions. Quantum machine learning, leveraging QC's power, can handle this complexity more efficiently. By integrating these models into traffic management systems, authorities can make more informed decisions promptly, improving traffic flow and transportation efficiency (Qu et al., 2022).

(iv) The urgent need for sustainable transportation solutions has led to a significant shift towards EVs in response to rising carbon dioxide emissions. However, this transition presents challenges, particularly in optimizing energy management systems and charging infrastructure. Enhancing energy management systems for EVs, charging infrastructure, and lithium-ion battery technology is critical for advancing electric transportation. By leveraging QAs such as quantum annealing and variational QAs, researchers can optimize energy usage in EVs and extend battery life (Dalyac et al., 2021; Luckow et al., 2021). Additionally, QC can enhance charging infrastructure design and optimization, leading to faster and more reliable charging solutions (Dalyac et al., 2021). Moreover, quantum simulations can improve lithium-ion battery performance, such as increasing energy density and charging rates (Delgado et al., 2022).

(v) As urbanization increases, the demand for efficient parking solutions intensifies, necessitating innovative approaches to optimize smart parking systems. Enhancing smart parking systems through QC involves various avenues. QAs like quantum annealing or optimization techniques can refine parking space allocation and optimization for vehicles as well as solar parking lots for EVs (Mladenović et al., 2021; Rahmani-andebili, 2016). Integrating quantum sensors enhances data accuracy and efficiency (Sureshkumar and Rajesh, 2018). Quantum communication protocols ensure secure data transmission, bolstering privacy and security (Malina et al., 2021). Lastly, quantum machine learning enables predictive analytics for parking demand forecasting, enhancing system efficiency. By embracing these strategies, smart parking systems can attain higher performance, reliability, and user experience.

(vi) The growing complexity of transportation systems demands innovative solutions for effective data collection and analysis to enhance operational efficiency and sustainability. Exploring quantum sensors and quantum-inspired sensor networks holds promise for advancing data collection and analysis in transportation systems. Quantum sensors offer heightened precision and sensitivity, improving traffic monitoring and environmental sensing. Embracing these technologies can revolutionize transportation systems by providing real-time insights and optimizing resource allocation for enhanced mobility and sustainability (Su et al., 2012; Marciniak et al., 2022).

(vii) Future researchers can enhance current research by integrating author co-citation network analysis and institution co-occurrence analysis. These methods provide insights into scholarly collaboration and institutional contributions, fostering deeper understanding and collaboration within the research community.

A generic quantum framework

Fig. 14 presents a comprehensive quantum framework that researchers can employ in their endeavors. This hybrid classical-quantum framework proves particularly advantageous within the context of the Noisy Intermediate-Scale Quantum era. Comprising five distinct stages, the framework facilitates a systematic approach to problem-solving. In the initial stage (Step 1), the model is supplied with input, which may encompass various forms of data, including sensor data, synthetic data, or publicly available datasets. Subsequently, in Step 2, known as quantum encoding or state preparation, a conversion from classical data to quantum data takes place. This transformation can be executed utilizing diverse encoding techniques such as Basic Encoding, Amplitude Encoding, Angle Encoding, encoding datasets by Hamiltonian, or data encoding as a Feature Map (Ghosh, 2021). Moving forward to Step 3, the quantum model undergoes evaluation tailored to the specific problem at hand. This evaluation may involve various quantum models, including "Quantum Neural Networks", "Quantum Convolutional Neural Networks", "QGA", "QAOA", and "Quantum Evolutionary Algorithms". Step 4 entails the conversion of quantum data back to classical data utilizing techniques such as Quantum Wavelet Transform (Mahmud et al., 2022). Following this conversion, the classical model is evaluated based on the requirements of the problem. The process involves the assessment of the cost function and the iterative updating of parameters until a predefined threshold is achieved. Finally, in Step 5, the output is generated, which consolidates the predictions made by the model. This structured approach facilitates efficient problem-solving within the QC domain while leveraging the capabilities of both classical and QC paradigms.

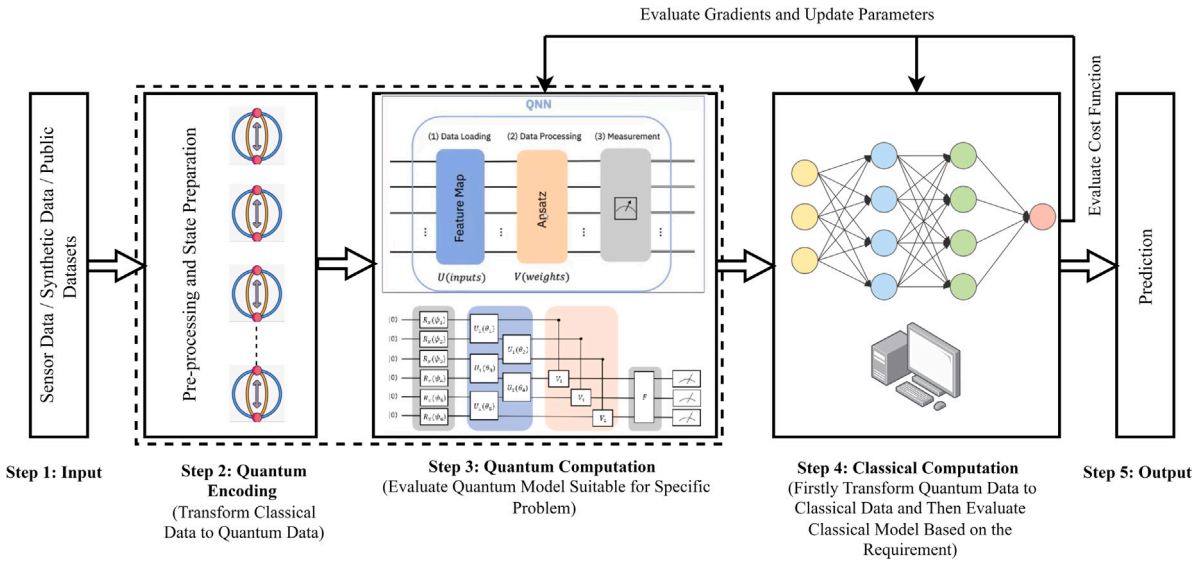


Fig. 14. A generic quantum framework.

8. Implications of the study

The implication of this study facilitates the advancement of QEITS research. Some of the implications of the study are given below.

(i) It provides an introductory source of Quantum Enhanced Intelligent Transportation System benefits globally, aiding new researchers in developing their interests and foundational knowledge.
(ii) It delineates primary research domains and subdomains, directing researchers towards novel avenues with potential practical applications.
(iii) Insights such as the identification of top contributing countries and influential active research areas offer valuable guidance to researchers, investors, and policymakers in facilitating development initiatives.
(iv) Moreover, the review article may assist developers by elucidating primary challenges, potential opportunities, future research directions, and generic quantum framework fostering the creation of innovative solutions.

9. Conclusions

The current paper acts as a comprehensive portrayal of the progress and trends within an ever-evolving scientific community, attained by employing a scientometric review method on QEITS. The exploration of four domains within the study facilitates effective communication and analysis among scientists, guiding future research directions. To offer a broader perspective, various analyses are conducted on each domain, including examination of publication patterns, assessment of significant contributing nations, investigation of DCoNs, and exploration of KCoNs. The notable findings within the realm of QEITS research reveal the prominence of AVCDS as a principal research frontier. Publications within the DVRN and EVTS domains have demonstrated substantial growth and elevated quality. Notably, the EVTS domain has swiftly evolved in the last year, indicating its promising potential as an emerging area. The identification of emerging trends and actively explored research territories within each domain is emphasized. The research offers an interesting opportunity for scientists and researchers to consider strategic approaches that can lead to substantial expansion and notable advancements in this domain. For future work, the authors will delve into the specified domains of QEITS to gain more comprehensive knowledge and understanding.

CRediT authorship contribution statement

Monika: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Formal analysis, Data curation. **Sandeep Kumar Sood:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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